CHRONO-Fair: A Controlled Validation Study of Time-Resolved Counterfactual Fairness Monitoring in Clinical Machine Learning

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Clinical machine learning models are now used across triage, risk scoring, and length-of-stay prediction. Documented subgroup-specific failures have emerged after release, and current auditing tools remain largely snapshot-based. The auditing tools in routine use respond to this by computing a group fairness metric on a fixed snapshot of predictions and returning a pass-or-fail verdict against a threshold such as the 0.8 rule. However, a point-estimate verdict hides whether the underlying number is stable. On the real Texas-100X discharge dataset (925 128 records), the worst-group true-positive-rate verdict for race reads "fair" at 0.824. The same verdict flips to "unfair" in 4 of 20 hospital-network resamples, a Verdict Flip Rate of 0.20. A batch audit cannot determine when, or whether, that verdict reversal recurs as new patients arrive.

To address this gap, we present CHRONO-Fair. The framework extends prior VFR-Audit-style static auditing into a post-release time-resolved monitoring layer. It watches a deployed clinical machine learning model after release. CHRONO-Fair does not perform pre-release mitigation. The monitored model may start from a passing point-estimate fairness verdict on every protected attribute, as in the Standard-to-Fair setup of prior VFR-Audit. The main method is a per-cell anytime-valid e-process on a counterfactual flip indicator, calibrated against a pre-deployment baseline flip rate ρ_0. A step-wise Benjamini-Hochberg layer is applied across monitored cells at each inspection time. Three supporting modules accompany the main method. Flip Hazard gives a group-stratified time-to-flip survival view. An entropy-based diagnostic attribution summarises whether a flip signal is more consistent with data-driven noise, model uncertainty, or a mixed pattern. RCAP, a Wasserstein-1 rank-allocation statistic, extends monitoring to length-of-stay regression. The evidence combines real Texas-100X batch verdict-stability analysis (925 128 records) with Texas-100X-calibrated controlled streaming and replay experiments. On the streams, the e-process detected an injected drift in 50 of 50 Monte-Carlo runs at a median delay of 828 patients, against 21 of 50 for ADWIN. The empirical false-alarm rate stayed at or below the nominal level for every tested α. We claim a controlled-validation result for the monitoring layer, not prospective clinical validation. Prospective evaluation on a live electronic-health-record stream remains the necessary next step before any operational adoption.

Additional Key Words and Phrases: algorithmic fairness, clinical machine learning, anytime-valid inference, fairness drift, post-deployment monitoring, length-of-stay prediction

1 INTRODUCTION

Machine learning models support clinical practice across sepsis early warning, length-of-stay forecasting, and discharge planning. Documented post-release fairness failures have accumulated alongside this adoption. Obermeyer et al. [83] traced racial bias in a widely used care-management algorithm to its cost-as-proxy label, and reported that correcting the label nearly tripled the share of Black patients flagged for additional care. Seyyed-Kalantari et al. [99] reported higher underdiagnosis rates for female and minority patients across multi-site chest-radiograph classifiers, with the largest gaps on intersectional subgroups. Habib et al. [45] and Wong et al. [115] found that the Epic Sepsis Model achieved an external-validation area under the receiver operating characteristic curve of 0.63, well below the vendor-reported range. Davis et al. [29] extended the picture along the time axis, showing across eleven years of VA cohort data that fairness gaps drift after release, often toward widening.

These post-release failures have pushed regulatory and reporting expectations from one-off model validation toward continuous lifecycle monitoring. FDA PCCP guidance [104] motivates predefined monitoring plans for model changes, including bias-related monitoring where relevant. EU AI Act [38] quality-management and post-market monitoring obligations motivate ongoing

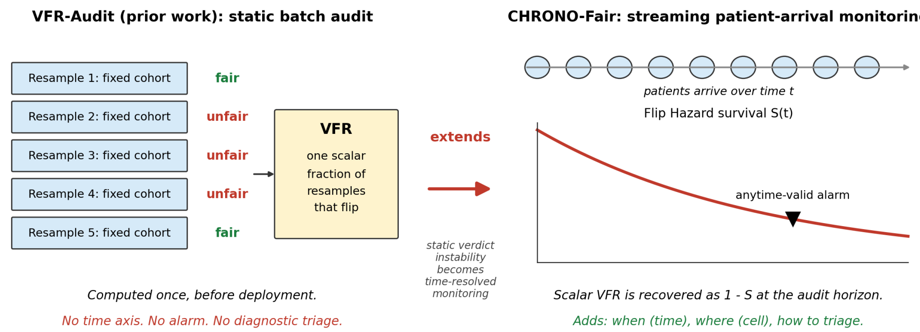

Fig. 1. VFR-Audit reports a static batch scalar. CHRONO-Fair extends the same verdict-flip object into time-resolved monitoring with an alarm and diagnostic triage.

bias surveillance for high-risk medical AI systems, with provisions binding from August 2027. The DECIDE-AI [106], TRIPOD+AI [26], and CONSORT-AI [70] reporting standards treat lifecycle evaluation as expected practice rather than as an extension. The clinical-informatics literature has begun to operationalise the same shift. You et al. [116] propose a clinical-trials-informed four-phase framework that places post-deployment monitoring as a co-equal phase with safety, efficacy, and effectiveness. Ansari et al. [8] survey the open challenges of postmarket surveillance for clinical prediction models. McCradden et al. [76] and Eaneff et al. [36] argue, in the same direction, for "algorithmic stewardship" as a regulatory category.

The toolchain available to satisfy that expectation, however, has not caught up. Aequitas, Fairlearn, AIF360, and the Microsoft Responsible AI Dashboard, the auditing packages in routine use, all operate in batch mode on a fixed set of predictions. Online fair-learning work [51, 52, 82] monitors a stream but has been demonstrated on COMPAS and credit data rather than on a clinical cohort. Anytime-valid inference [44, 49, 93, 114] supplies the sequential statistical machinery, yet it has been applied to A/B-test monitoring and calibration drift rather than to fairness drift at intersectional granularity. Predictive-multiplicity work [13, 27, 74] characterises pipeline instability but frames it as reliability rather than fairness. These lines of work generally do not report when a thresholded fairness verdict, the object hospitals actually act on, begins to flip as patients arrive.

The closest precedent on the verdict-stability question is the prior VFR-Audit work [7], which introduced the Verdict Flip Rate, a scalar that quantifies how often a fairness verdict flips between resamples of a hospital-network bootstrap. It captures something the canonical group metrics miss, the instability of the verdict itself, and it does so on the pass-or-fail object that governance teams act on. It does so, however, as one batch number at one threshold, and it is silent on when, where, or why a verdict moves after release. The present paper extends that line along the time axis.

Contribution. The framework introduced in this paper, CHRONO-Fair, sits above a deployed clinical machine learning model and monitors whether its thresholded fairness verdict continues to hold as patients arrive. The deployed model may be uncorrected or may have been fairness-adjusted by any pre-release mitigation method; CHRONO-Fair is agnostic to that choice. This paper makes four contributions, evaluated through real Texas-100X batch evidence and Texas-100X-calibrated controlled streaming and replay experiments. The main contribution is a per-cell e-process on a counterfactual flip indicator, calibrated against a pre-deployment baseline flip rate and combined with an inspection-time cross-cell adjustment. Three supporting modules give the governance-review interface its evidence. The first is a Flip Hazard survival view of group-stratified time-to-flip. The second is an entropy-based diagnostic attribution that summarises whether a detected flip signal is more consistent with data-driven noise, model uncertainty, or a mixed pattern. The third is a Wasserstein-1 rank-allocation statistic for length-of-stay regression. A dashboard and an automatically generated inspection report bundle the four outputs for a clinical safety officer. The novelty is not a new survival estimator or a new e-process. The contribution is the integration of time-resolved flip-hazard estimation, anytime-valid per-cell alarm evidence, diagnostic flip attribution, and rank-allocation regression monitoring around the governance object that hospitals act on: the thresholded fairness verdict.

2 RELATED WORK

2.1 Group, individual, and counterfactual fairness

Dwork et al. [35] defined individual fairness as a Lipschitz constraint. Two individuals within distance d on a task metric must receive outcome distributions within distance D. Hardt et al. [46] defined equalised odds as equal true-positive and false-positive rates across groups. Equal opportunity relaxes this to equal true-positive rate only. Chouldechova [24] proved that, when group base rates differ, a classifier cannot hold predictive parity, equal false-positive rate, and equal false-negative rate at once. Kleinberg et al. [64] proved a parallel result: calibration and class-balanced error are jointly attainable only under equal base rates or perfect prediction. Pleiss et al. [88] showed the conflict persists for any non-trivial calibration constraint and quantified the achievable relaxation. Kusner et al. [66] defined counterfactual fairness as invariance of the prediction distribution under a do-intervention on the protected attribute in a structural causal model. Chiappa [23] restricted this to unfair causal paths only, leaving resolving paths intact. Nabi and Shpitser [79] constrained path-specific effects during estimation by a constrained likelihood. Kilbertus et al. [62] separated proxy discrimination from explanatory effects in a causal graph. Makhlouf et al. [72] mapped 19 causal fairness notions to the data assumptions each one needs. The surveys of Verma and Rubin [107], Mehrabi et al. [77], Pessach and Shmueli [86], and Caton and Haas [19] catalogue 20 or more metrics. They report that the choice of metric, not the model, drives most reported disparity. On mitigation, Agarwal et al. [2] reduced fair classification to a sequence of cost-weighted subproblems solved by a saddle-point game. Donini et al. [32] added a fairness term to empirical risk minimisation with a generalisation bound. Calmon et al. [17] and Kamiran and Calders [58] reweighted or transformed training data before fitting. Madras et al. [71] and Lahoti et al. [67] learned representations that an adversary cannot use to recover the protected attribute. Wang et al. [112] bounded fairness violation when group labels are noisy. Kallus et al. [57] estimated disparity by combining a main dataset with an auxiliary dataset that carries the protected attribute.

2.2 Subgroup, multicalibration, and statistical-audit perspectives

Kearns et al. [60] showed that parity on coarse groups can hide large violations on structured subgroups, an effect they named fairness gerrymandering. Hébert-Johnson et al. [48] defined multicalibration as calibration that holds simultaneously on every subgroup in a computationally identifiable class. Kim et al. [63] produced multiaccuracy by post-processing, auditing for subgroups where the residual correlates with subgroup membership. Hashimoto et al. [47], Sagawa et al. [95], and Duchi and Namkoong [33] minimised worst-group loss through distributionally robust optimisation over an uncertainty set of group mixtures. Martínez et al. [73] characterised the accuracy-fairness frontier as a Pareto set and computed it through a multi-objective descent. Cherian and Candès [22] produced finite-sample confidence intervals for a fairness metric using bootstrap and rank-based statistics. Besse et al. [11] reported that the sampling variance of statistical parity can exceed the disparity itself when subgroup size is below a few hundred. Coston et al. [28] evaluated counterfactual risk under selective labels through inverse-propensity weighting. These works audit a fixed snapshot. None updates a fairness verdict sequentially as patients arrive.

2.3 Predictive multiplicity, model multiplicity, and stability

Breiman [16] observed that distinct models can attain near-identical loss while disagreeing on individuals. Marx et al. [74] formalised this as predictive multiplicity. They defined ambiguity as the fraction of inputs that receive a conflicting prediction from some model within an ϵ-loss Rashomon set. Watson-Daniels et al. [113] extended the measure to probabilistic outputs through the range of predicted scores. Semenova et al. [98] linked a large Rashomon set to the existence of a simpler accurate model. Dong and Rudin [31] computed variable-importance ranges across the Rashomon set rather than for one model. Black et al. [13] argued that model multiplicity gives a designer latitude to select a fairer model at no accuracy cost. Cooper et al. [27] showed that a measured group gap can be a variance artefact: averaging predictions across the Rashomon set shrank reported disparity. Miller and Blume [78] defined the empirical Decision Flip Rate as the fraction of patients whose binary decision changes between bootstrap re-trainings. Black et al. [14] detected individual discrimination through an optimal-transport map between group-conditional input distributions. The Verdict Flip Rate of [7] measured how often the fairness verdict, not the prediction, flips between hospital-network resamples.

2.4 Uncertainty quantification and its decomposition

Depeweg et al. [30] decomposed predictive uncertainty into two terms. Aleatoric uncertainty is the mean of per-model entropies. Epistemic uncertainty is the entropy of the mean prediction minus that mean, equal to the mutual information between label and parameters. Hüllermeier and Waegeman [50] formalised the distinction and noted that only the epistemic term shrinks with more data. Lakshminarayanan et al. [68] estimated epistemic uncertainty from the disagreement of M independently trained networks. Gal and Ghahramani [41] approximated the same quantity through Monte Carlo dropout. Kendall and Gal [61] separated learned input-dependent variance from model uncertainty in a single loss. Ovadia et al. [84] reported that deep ensembles kept calibration error lowest among uncertainty methods under distribution shift. Alghamdi et al. [4] applied the aleatoric-epistemic split to discrimination, attributing the irreducible part to the data and the reducible part to the model class. Romano et al. [94] and Gibbs and Candès [43] produced prediction sets with a finite-sample coverage guarantee, the latter adapting the set width online under shift. Barber et al. [9] extended conformal coverage to non-exchangeable data with a bound that degrades in the total-variation distance from exchangeability. Angelopoulos and Bates [6] surveyed the coverage guarantees that the conformal family provides. Dutta et al. [34] and Calmon et al. [18] decomposed discrimination through partial information, separating the exempt component from the unfair component.

2.5 Anytime-valid inference, drift detection, online learning

Ville [108] proved that a non-negative supermartingale started at one crosses $1/\alpha$ with probability at most α. This single inequality underwrites every e-process. Wald [111] built the sequential probability ratio test, which stops as soon as a likelihood ratio leaves a fixed interval. Howard et al. [49] produced confidence sequences, intervals that cover the parameter simultaneously at every sample size with probability $1 - \alpha$. Waudby-Smith and Ramdas [114] estimated a bounded mean by betting. The wealth process $\prod_t (1 + \lambda_t(X_t - m))$ is a martingale under the null mean m. The GROW rule picks λ_t to maximise expected log-growth. Ramdas et al. [93] and Shafer

[100] unified these results as game-theoretic statistics, where evidence equals the capital won against the null. Grünwald et al. [44] characterised the e-variable that is growth-optimal against a composite alternative. Johari et al. [55] applied always-valid p-values to continuous A/B-test monitoring, removing the penalty for repeated peeking. Podkopaev and Ramdas [89] tracked a deployed model's risk through a betting martingale on the loss sequence. Foster and Stine [40] introduced alpha-investing, which spends a false-discovery budget across an ordered hypothesis stream. Javanmard and Montanari [53] produced the LORD rules that control online false-discovery rate with a decaying memory. Bifet and Gavalda [12] introduced ADWIN, which keeps a variable-length window and drops the older half when two sub-window means differ by a Hoeffding bound. Gama et al. [42] taxonomised concept drift into sudden, gradual, and recurring types. Rabanser et al. [91] reported that dimensionality reduction followed by a two-sample test detected dataset shift most reliably. Iosifidis and Ntoutsi [51, 52] maintained a fairness constraint online by reweighting the loss as a stream evolves. We are not aware of prior work that combines counterfactual flip monitoring with per-cell anytime-valid control and an inspection-time cross-cell adjustment.

2.6 Clinical machine learning fairness, prediction, and deployment

Obermeyer et al. [83] traced the bias of a care-management algorithm to its label. The model predicted healthcare cost as a proxy for healthcare need. At a fixed risk score, Black patients carried more chronic conditions than White patients. Correcting the label raised the share of Black patients flagged for extra care from 17.7% to 46.5%. Vyas et al. [109] catalogued race-adjusted clinical formulas, including an eGFR correction that lowered estimated kidney disease severity for Black patients. Seyyed-Kalantari et al. [99] reported that chest radiograph classifiers had higher false-negative rates for female, Black, and publicly insured patients, with the largest underdiagnosis on intersectional subgroups. Wong et al. [115] externally validated the Epic Sepsis Model. They found an area under the receiver operating characteristic curve of 0.63, against the vendor-reported 0.76 to 0.83. Habib et al. [45] confirmed the external-validation gap and the high alert burden. Noseworthy et al. [81] tested an electrocardiogram deep model and found stable performance across race once age and sex were matched. Larrazabal et al. [69] showed that a sex-imbalanced training set lowered diagnostic accuracy for the under-represented sex. Coley et al. [25] reported that a suicide-risk model had lower sensitivity for Black and American Indian patients. Adam et al. [1] found that race-correlated language in clinical notes shifted downstream model recommendations. Pierson et al. [87] trained a pain model on patient-reported scores rather than radiologist labels and explained a larger share of the racial pain gap. Davis et al. [29] tracked an in-use model across eleven years of VA data and observed subgroup performance drift after deployment. Rajkomar et al. [92] and Chen et al. [20] set out fairness desiderata for clinical machine learning. Singh et al. [101] argued that a deployed clinical model needs continual rather than one-off evaluation. Finlayson et al. [39] catalogued the dataset-shift modes that degrade clinical models, including case-mix and practice-pattern change. Vasey et al. [106] and Collins et al. [26] issued the DECIDE-AI and TRIPOD+AI reporting standards, both of which now require subgroup performance reporting. Sculley et al. [97] and Breck et al. [15] documented the monitoring debt that accrues when a model ships without a test harness. The MIMIC-IV [56] and eICU [90] databases supply the public critical-care streams used for external validation. Texas-100X [54] provides 925 128 hospital discharge records with the demographic mix this paper's synthesiser reproduces.

2.7 Counterfactual auditing of deployed classifiers

Maughan and Near [75] proposed prediction sensitivity, a per-batch scalar. It estimates how often a counterfactual swap of the protected attribute changes the verdict, without the attribute at inference time. Wachter et al. [110] defined a counterfactual explanation as the smallest input

Table 1. Fairness auditing dashboards and monitoring tools, against the operating regime a clinical post-deployment monitor needs. Entries cover function, operating mode, output object, and the residual gap that CHRONO-Fair targets. No new statistical claim is made by this table; it consolidates feature descriptions from each tool's public documentation and from the cited drift- and verdict-stability work.

Tool or work	Main function	Operating mode	Output object	Residual gap for Cl
Aequitas	Group-rate and error-rate parity ratios per protected attribute	Batch on a fixed prediction file	Per-attribute bias report (web table)	No sequential alarm anytime-valid guar
Fairlearn	Selection-rate, equalised-odds, demographic-parity metrics; reduction-based mitigator	Batch on held-out evaluation set	Metric dashboard plus mitigator object	No streaming verd nostic split
AIF360	Library of 20 group and individual metrics with pre-, in-, post-processing mitigators	Batch on held-out evaluation set	Metric report and corrected model	Inspection-time s flip-monitoring or mate
Microsoft RAI Dashboard	Fairlearn + InterpretML + counterfactual-explanation widgets in one developer view	Batch in notebook session	Interactive review widget	Aimed at develope at continuous gove
Fairness drift on VA cohort [29]	Quarterly subgroup performance tracking on deployed VA model	Periodic batch (quarterly)	Trend report per subgroup	No anytime-valid g gle metric per cell
Verdict Flip Rate [7]	Static verdict-stability scalar on Texas-100X batch	Static (one batch evaluation)	Single VFR number per protected attribute	No time axis; no pe tial alarm

change that flips the decision. Ustun et al. [105] certified whether such a change is attainable through actionable features only. Chen et al. [21] bounded a disparity estimate when the protected attribute is unobserved at audit time. These methods return a scalar or a per-instance explanation at a fixed time. They provide no anytime-valid alarm and no diagnostic decomposition of the flip into a data signal and a model signal. CHRONO-Fair targets that gap.

2.8 Fairness auditing dashboards and post-deployment monitoring tools

A separate line of work has packaged the fairness audit as a tool that an analyst or governance team can run. Aequitas, released by Saleiro and colleagues, exposes group-rate, false-positive-rate, and false-negative-rate parity ratios across protected groups on a fixed prediction file and renders the resulting bias report as a per-attribute web table. Fairlearn, originating from the Microsoft FairLearn project of Bird and colleagues, ships a Python interface that computes selection-rate, equalised-odds, and demographic-parity statistics on a held-out evaluation set together with an in-processing reduction-to-classification mitigator. AIF360, released by Bellamy and colleagues at IBM Research, bundles roughly twenty group and individual fairness metrics with pre-, in-, and post-processing mitigators in one library. The Microsoft Responsible AI Dashboard composes Fairlearn, InterpretML, error-analysis, and counterfactual-explanation components inside a single notebook-embeddable widget aimed at a model developer's review session. All four artefacts operate in batch on a fixed set of predictions, present a snapshot at the moment of inspection, and offer no anytime-valid statistical guarantee under repeated peeking. The Davis et al. [29] VA fairness-drift study sits on the empirical side of the same gap. It tracks subgroup performance quarterly across eleven years of deployed predictions. It demonstrates that the drift is real. It stops short of providing a per-cell sequential alarm at intersectional granularity. The Verdict Flip Rate of [7] is the closest prior artefact to the present work, in that it scores the stability of the fairness verdict itself rather than of a point metric; it does so in batch mode with no time axis. Table 1 summarises this auditing-tool layer along five axes that matter for clinical post-deployment use.

Takeaway 1. The four auditing toolkits in routine use (Aequitas, Fairlearn, AIF360, Microsoft RAI Dashboard) all operate in batch on a fixed prediction file. None carries a sequential statistical guarantee under repeated inspection.

Takeaway 2. The closest streaming clinical evidence, the quarterly VA fairness-drift study of Davis et al., demonstrates that the drift is real but stops at quarterly aggregation and at a single subgroup metric. It leaves open the per-cell, anytime-valid layer that CHRONO-Fair contributes.

Takeaway 3. The Verdict Flip Rate of the prior VFR-Audit work is the closest object to CHRONO-Fair's monitored quantity, but it is reported as a single batch scalar with no time axis. CHRONO-Fair lifts the same quantity into a per-patient hazard curve under an anytime-valid alarm rule.

Takeaway 4. Existing tools report the current disparity. They do not report when it opened, whether the signal exceeds noise under repeated peeking, or whether a data-pipeline issue or a sample-size issue is more consistent with the data. CHRONO-Fair is built around those three questions.

Each CHRONO-Fair module corresponds to one of the gaps identified above. The Flip Hazard estimator closes the time-axis gap left open by the four batch toolkits and by static VFR, by reporting a per-arrival-index survival curve rather than a scalar. The per-cell shrunk-GROW e-process with step-wise Benjamini-Hochberg adjustment closes the peeking-penalty gap left open by every batch tool and by the quarterly Davis et al. cadence, by giving a per-inspection-time guarantee under Ville's inequality. The aleatoric-epistemic decomposition closes the interpretation gap that all of Aequitas, Fairlearn, AIF360, and the Microsoft RAI Dashboard leave open, by attaching a diagnostic triage label (data-side, model-side, or mixed) to every alarmed cell rather than reporting only its magnitude. RCAP closes the regression-output gap that the flip-rate and verdict literature has not addressed, by quantifying counterfactual rank-position shift on a length-of-stay output for which a binary verdict is not defined. This four-way mapping summarises how the supporting modules align with the literature gaps the paper targets.

2.9 Positioning against prior work

Table 2 places CHRONO-Fair against eight families of prior work on nine capability axes. The axes are the ones a post-deployment clinical monitor needs. Static or streaming records the operating mode. Verdict reliability records whether the method reports the stability of a pass or fail verdict. Anytime-valid per-cell monitoring records a false-alarm rate bounded under inspection at every step. Cross-cell adjustment records a multiple-testing correction across monitored cells. Intersectional records cells defined by two or more protected attributes. Counterfactual-flip monitoring records tracking of verdict flips under a protected-attribute counterfactual. Diagnostic triage output records whether the framework attributes a detected gap to a likely data, model, or mixed signal source. Regression support records handling of a continuous, rank-allocation output. Real clinical evidence records the empirical clinical data used.

Read row by row, the comparison suggests a recurring gap across existing method families, since no surveyed family in this table holds more than two of the nine capabilities at once. The nearest entries each fall short on a different axis. Anytime-valid risk tracking [89] is streaming and anytime-valid, but it monitors aggregate loss rather than a group-conditional fairness verdict and offers no diagnostic attribution. Fairness-drift monitoring [29] is the closest clinical work, yet it operates on a quarterly cadence, monitors a single metric, and gives no statistical guarantee under repeated inspection. The Verdict Flip Rate [7] measures verdict instability directly but collapses it to a scalar with no time axis. CHRONO-Fair occupies the remaining position by combining the

Table 2. Capability comparison of fairness-audit method families. “Y” denotes the capability is present as a designed feature, “–” denotes absent or not reported, and “P” denotes partial support. The final row is the contribution of this paper; the footnote states what each “P” on that row means.

Method family (representative work)	Static/ streaming	Verdict reliability	Anytime-valid per-cell	Cross-cell adjustment	Inter-sectional	CF-flip monitoring	Diagnostic attribution	Regression/ rank-alloc.	Real clinical evidence
Group metrics, batch toolkits [46]	static	–	–	–	P	–	–	–	various
Bootstrap-CI fairness audit [22]	static	Y	–	–	–	–	–	–	–
Predictive multiplicity, eDFR [74, 78]	static	Y	–	–	–	–	–	–	some
Verdict Flip Rate [7]	static	Y	–	–	P	Y	–	–	Texas-100X batch
Prediction sensitivity [75]	per batch	–	–	–	–	Y	–	–	–
Online fair learning [51]	streaming	–	–	–	–	–	–	–	–
Fairness-drift monitoring [29]	quarterly	P	–	–	P	–	–	–	VA cohort
Anytime-valid risk tracking [89]	streaming	–	Y	–	–	–	–	–	–
CHRONO-Fair (this paper)	**streaming**	**Y**	**Y**	**P**	**Y**	**Y**	**P**	**Y**	**P**

Footnote. For the CHRONO-Fair row, “P” on cross-cell adjustment means an inspection-time step-wise Benjamini-Hochberg adjustment rather than a full time-uniform online false-discovery procedure. “P” on diagnostic attribution means an aleatoric/epistemic triage signal, not a confirmed causal root cause. “P” on real clinical evidence means a real Texas-100X batch verdict-stability analysis combined with Texas-100X-calibrated controlled streaming and replay experiments, not prospective deployment evidence.

streaming, anytime-valid, intersectional, and diagnostic capabilities, while the three partial marks in its row record exactly where its support is qualified rather than complete.

2.10 Open gaps addressed by this paper

Four gaps follow from Table 2. *Gap 1, the time axis.* A batch verdict cannot say when a fairness gap opened during deployment, and a clinical response needs that timestamp. *Gap 2, the peeking penalty.* A governance team inspects a monitor often. A fixed-sample test inflates its false-alarm rate under repeated looks. A continuous monitor therefore needs an anytime-valid guarantee. *Gap 3, the diagnostic interpretation.* A detected gap with no diagnostic interpretation does not tell a team whether to fix the data pipeline or to collect more samples. *Gap 4, regression outputs.* Length-of-stay and cost models produce continuous outputs. The flip-rate and verdict literature is built for binary classification. CHRONO-Fair addresses the four gaps with the Flip Hazard estimator, the e-process, the aleatoric-epistemic decomposition, and RCAP, in that order.

3 PROBLEM FORMULATION

3.1 Clinical fairness, the fairness verdict, and the flip indicator

This paper uses three terms with fixed meaning throughout. *Clinical fairness* denotes the property that the thresholded decisions of a deployed clinical machine learning model do not depend on a patient’s legally or ethically protected attributes beyond what is clinically justified. Compliance is evaluated by an agreed group-fairness metric and threshold (for example disparate impact above 0.8). The *fairness verdict* is the binary pass-or-fail outcome of that evaluation at a given threshold. The *counterfactual verdict flip*, or *verdict flip* for short, is the event in which the verdict changes between the model’s factual decision on a patient and its counterfactual decision under a swap of the protected attribute. The framework treats the verdict flip as the observable governance object and monitors its rate as patients arrive.

3.2 Notation and assumptions

Let $\{(X_i, A_i, Y_i)\}_{i\geq 1}$ be a stream of patient records, where $X_i \in \mathbb{R}^d$ are non-sensitive features, A_i is a protected attribute taking values in a finite set $\mathcal{A}$, and $Y_i \in \{0, 1\}$ or $Y_i \in \mathbb{R}_{\geq 0}$ is the label. The arrival index t orders patients by admission time. The deployed model at index t is $f_t\colon (X, A) \to \hat{Y}$. The model may change between two indices through periodic retraining, threshold recalibration, or hot-fixes. Re-training events are exposed to CHRONO-Fair as version stamps. Between two retrains, the model is held fixed and the framework monitors the decision stream rather than the model parameters.

Let A_i^{cf} denote a counterfactual sensitive-attribute value drawn from $\mathcal{A}$. In this paper, A_i^{cf} is fixed to a reference level (typically the majority group). The counterfactual prediction $\hat{Y}_i^{\mathrm{cf}} = f_t(X_i, A_i^{\mathrm{cf}})$ is obtained by setting the protected attribute to its counterfactual value and removing the marginal effect on observable features through a feature-shift adjustment. This corresponds to the deployment-time counterfactual of Maughan and Near [75]. It is weaker than the SCM-based counterfactual of Kusner et al. [66] but it is computable at inference time without an audited causal graph.

3.3 Flip indicator

For binary classification with threshold τ, the flip indicator is

$$Z_i = \mathbf{1}\left\{\mathbf{1}\left\{\hat{Y}_i \geq \tau\right\} \neq \mathbf{1}\left\{\hat{Y}_i^{\mathrm{cf}} \geq \tau\right\}\right\}. \tag{1}$$

For regression, a threshold-free analogue is the counterfactual rank shift formalised in Section 4.4. The flip indicator depends on f_t at the patient's arrival index, not on f_t at the reporting time, so retraining events do not retroactively invalidate prior Z_i.

3.4 Baseline calibration

Each cell c defined by a value of the protected attribute has a baseline flip rate $\rho_0^{(c)}$. The baseline is calibrated on the pre-deployment validation set used to lock the model for release. The validation set should match the deployment population and contain at least 500 patients per cell. If the per-cell calibration size falls below 500, $\rho_0^{(c)}$ is set to the marginal flip rate on the same validation set and a wider warm-up window is used. Calibration is repeated each time the model is replaced.

4 METHOD

CHRONO-Fair is one framework with four coordinated estimators that share a single counterfactual flip indicator. Figure 2 shows the data flow. A patient stream feeds the deployed model and its counterfactual, the flip indicator is computed, and the four estimators consume it. The Flip Hazard estimator and the e-process are sequential. The decomposition and RCAP run on demand. The Inspector report aggregates the four outputs for governance review.

4.1 Flip Hazard estimator

Definition 4.1 (Flip Hazard). For protected group a, the Flip Hazard function is

$$\lambda_a(t) = \lim_{\Delta \to 0} \frac{1}{\Delta} \Pr(Z_t = 1, T \geq t \mid A = a) \big/ \Pr(T \geq t \mid A = a), \tag{2}$$

where $T = \inf\{t : Z_t = 1\}$ is the first flip time and the limit is taken on the patient-arrival index.

The no-flip survival is $S_a(t) = \exp(-\int_0^t \lambda_a(s)\, ds)$. The scalar VFR of [7] corresponds to the aggregate $1 - S_a(\tau^\star)$ at the full audit horizon $\tau^\star$. The framework reports survival as a curve, not a scalar.

The estimator is non-parametric, so it makes no distributional assumption on the flip-time. Kaplan-Meier survival [59] and Nelson-Aalen cumulative hazard [80] are computed per group, and Greenwood confidence bands are produced around each survival curve. The pairwise log-rank test

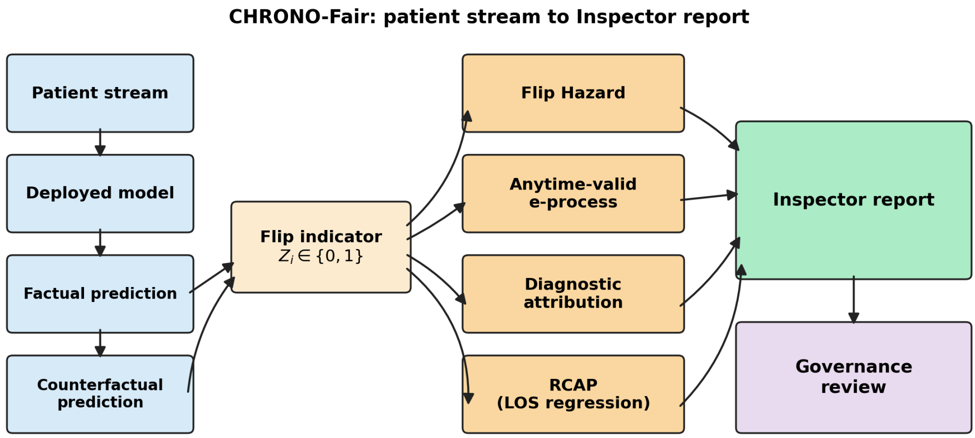

Fig. 2. CHRONO-Fair architecture. The patient stream feeds the deployed model, the factual and counterfactual predictions yield the flip indicator, and four modules consume it: Flip Hazard survival, the per-cell anytime-valid e-process, diagnostic attribution, and RCAP. The Inspector report aggregates the four outputs for governance review.

[5] is applied for between-group comparisons. The Restricted Mean Flip Time at horizon $\tau^\star$ is

$$\mathrm{RMFT}_a(\tau^\star) = \int_0^{\tau^\star} S_a(t)\,dt, \tag{3}$$

and admits a closed-form step-integration on the empirical KM curve.

4.2 Anytime-valid e-process

The e-process for one cell c tests $H_0\colon \mathbb{E}[Z_t] = \rho_0^{(c)}$ against the one-sided alternative $\mathbb{E}[Z_t] > \rho_0^{(c)}$:

$$E_t^{(c)} = \prod_{s=1}^{t} \big(1 + \lambda_s^{(c)}(Z_s - \rho_0^{(c)})\big), \tag{4}$$

where $\lambda_s^{(c)}$ is a predictable bet sequence measurable in $\sigma(Z_1, \ldots, Z_{s-1})$. The bet is clipped to $[-1/(1-\rho_0), 1/\rho_0]$ to guarantee $E_t^{(c)} \geq 0$. The framework uses a shrunk GROW rule:

$$\lambda_s = c\,\frac{\bar{Z}_{s-1} - \rho_0}{\rho_0(1-\rho_0)} \cdot \mathbf{1}\{s \geq n_{\mathrm{warm}}\}\,, \tag{5}$$

with shrinkage $c = 0.25$ and warm-up $n_{\mathrm{warm}} = 100$. The shrinkage and warm-up are conservative choices that trade detection speed for empirical false-alarm control; they are not load-bearing for anytime validity, only for finite-sample calibration on small streams.

Theorem 4.2 (Anytime-valid Type-I control). *Under H_0, the process $(E_t^{(c)})_{t\geq 0}$ is a non-negative supermartingale with $\mathbb{E}[E_t^{(c)}] \leq 1$. By Ville's inequality, for any stopping time τ,*

$$\Pr(\sup_{t\leq\tau} E_t^{(c)} \geq 1/\alpha \mid H_0) \leq \alpha. \tag{6}$$

The proof follows the supermartingale construction in Ramdas et al. [93], applied to the bounded random variable $Z_s - \rho_0 \in [-\rho_0, 1-\rho_0]$.

Intersectional multiple-testing control. For m disjoint cells, the framework runs m parallel monitors. At each inspection step, step-wise Benjamini-Hochberg [10] is applied to the e-value-implied p-values $p_t^{(c)} = 1/E_t^{(c)}$ clipped to $[0,1]$.

The e-process provides anytime-valid Type-I control for each monitored cell. The step-wise Benjamini-Hochberg layer is applied across cells at each inspection time and should not be interpreted as a time-uniform online false-discovery guarantee. The per-cell anytime-valid Type-I control of Theorem 4.2 holds for each cell separately. The cross-sectional step-wise Benjamini-Hochberg adjustment controls the false-discovery proportion among the cells inspected at a given step, under exchangeability of cell e-values. Extending CHRONO-Fair with a time-uniform online false-discovery procedure, such as LORD [53], SAFFRON, alpha-investing [40], or e-BH, is future work. The framework therefore reports findings as “flagged at inspection step t” rather than “time-uniformly significant”.

Complexity. Per arriving patient, the e-process update is $O(1)$ per cell. The step-wise BH sort across m cells is $O(m \log m)$. For a four-attribute deployment, m ranges from 13 marginal cells to 80 full intersectional cells. The per-patient update path was profiled on a 10 000-patient stream with 13 marginal cells. The measured latency was below 10 ms on a single CPU core. Ensemble decomposition and dashboard rendering were excluded from the profile.

4.3 Aleatoric and epistemic decomposition

Given a bootstrap ensemble of K classifiers with member probabilities $\{p_k(x)\}_{k=1}^K$ and counterfactual member probabilities $\{p_k^{\mathrm{cf}}(x)\}_{k=1}^K$, the per-instance flip mass is

$$\bar{p}^{\mathrm{flip}}(x) = \frac{1}{K}\sum_{k=1}^{K} \mathbf{1}\left\{(p_k(x) \geq \tau) \neq (p_k^{\mathrm{cf}}(x) \geq \tau)\right\}. \tag{7}$$

The total predictive entropy admits the decomposition of Depeweg et al. [30] into

$$H[\bar{p}] = \underbrace{\mathbb{E}_k H[p_k]}_{\text{aleatoric}} + \underbrace{\left(H[\bar{p}] - \mathbb{E}_k H[p_k]\right)}_{\text{epistemic}}. \tag{8}$$

For each instance, the flip mass is apportioned in proportion to the aleatoric and epistemic shares of the total entropy. Aggregated over a cell, the report summarises whether the flip signal is more consistent with data-driven noise (ensemble members that are individually confident and agree), with model uncertainty (members that disagree on the verdict), or with a mixed pattern (intermediate shares). The decomposition is intended as a diagnostic triage signal for governance review rather than as a causal attribution. Section 6.3 validates the formula on controlled inputs (Track 1) and reports the mixed-pattern behaviour of realistic ensembles (Track 2). A mixed triage label is the appropriate output for a real pipeline in which both data and model effects coexist. The mixed label is therefore not evidence of failure.

4.4 Regression Counterfactual Allocation Parity

Let $f(x,a)$ be a regressor producing a continuous output, length of stay in days in this paper. Let $\hat{F}_{\hat{Y}|A=a}$ be the empirical cumulative distribution function of the predicted output within group a. For patient i in group a, the predicted-rank position under the factual and the counterfactual attribute is

$$u_i(a) = \hat{F}_{\hat{Y}|A=a}\big(f(x_i, a)\big), \tag{9}$$

$$u_i(a') = \hat{F}_{\hat{Y}|A=a'}\big(f(x_i, a')\big). \tag{10}$$

Each u_i lies in $[0, 1]$ and gives the patient's percentile position in the group-conditional predicted-output distribution. The per-patient rank shift is $\Delta_{\text{rank}}(x_i, a, a') = u_i(a) - u_i(a')$. The aggregate RCAP statistic is the Wasserstein-1 distance between the two rank-position distributions

$$\text{RCAP}(a, a') = W_1\big(\{u_i(a) : i \in a\},\ \{u_i(a') : i \in a\}\big). \tag{11}$$

A bootstrap 95% confidence interval is computed by resampling the per-patient pairs $(u_i(a), u_i(a'))$ $B = 1000$ times.

RCAP measures how far a patient's predicted length-of-stay rank position shifts under the protected-attribute counterfactual. The rank position, not the raw error alone, determines prioritisation in length-of-stay-based bed allocation. The group-MAE gap collapses a directional, distribution-aware allocation shift into the average regression error and does not register an allocation-only change. RCAP is reported on the $[0, 1]$ rank scale, so a value of 0.0163 corresponds to a shift of 1.63 percentile points.

4.5 Inspector report

When the intersectional monitor flags a cell, the framework emits a structured report. Fields include: cell label, observed n, current flip rate, baseline flip rate, e-value, hazard ratio against baseline, aleatoric and epistemic shares, dominant component, recommended action, and a suggested regulatory mapping. The recommended action is derived deterministically from the dominant component: an aleatoric-dominated cell triggers a label-pipeline audit recommendation; an epistemic-dominated cell triggers a recommendation to enlarge the stratum sample. The regulatory mapping is a non-binding cross-reference to FDA PCCP [104] sections, EU AI Act [38] articles, and STANDING Together [3] recommendations, intended to accelerate review by clinical governance staff. Formal regulatory validation is future work.

4.6 Relationship to VFR-Audit and monitored model setup

CHRONO-Fair is not a fairness mitigator. It is a post-deployment monitoring framework. It can monitor any deployed clinical machine learning model after a pre-release fairness audit. The deployed model may be released without mitigation, or after reweighing, threshold adjustment, in-processing constraints, or any other pre-release method. In this paper, VFR-Audit [7] is cited only as the prior static audit framework and conceptual predecessor. VFR-Audit measures whether thresholded fairness verdicts are reliable under three sources of variation. The three sources are resampling, audit-size change, and cross-hospital site change. VFR-Audit introduces the Verdict Flip Rate as a static scalar summary of that reliability. CHRONO-Fair extends that idea by monitoring whether counterfactual fairness-verdict flips emerge over time after deployment.

The streaming experiments in Section 6 use a Fair monitored model adopted after the pre-release static audit reported in [7]. The Standard/Fair pair defines the monitored-model setup for the streaming experiments. It is not a named contribution of this paper. The monitored-model setup is reported in Section 6.10, where the Standard-versus-Fair numbers are tabulated.

CHRONO-Fair does not require any specific pre-release mitigation method. It monitors the stream of factual and counterfactual predictions produced by the deployed model. The same monitor therefore applies after reweighing, threshold adjustment, in-processing mitigation, or no mitigation at all. The baseline flip rate ρ_0 in Section 3 absorbs whichever pre-release configuration the deployment site used. The framework is in that sense agnostic to the upstream pipeline.

5 DEPLOYMENT

A proof-of-concept research prototype will accompany the manuscript to support reproducibility and inspection of the monitoring workflow, subject to anonymisation, institutional approval, and journal artifact-release requirements.

5.1 Software stack

The reference implementation uses the standard scientific Python stack (NumPy, pandas, scikit-learn, SciPy, matplotlib, statsmodels) with a Streamlit dashboard layer. No external API calls are made on the critical path. Exact package versions, random seeds, and configuration files accompany the reproducibility artifact rather than being treated as methodological contributions.

5.2 Data interface

The framework consumes a stream of records carrying an arrival timestamp, a patient identifier, the factual and counterfactual predicted labels, the optional predicted regression value, the protected attributes, and an optional model version stamp. A reference adapter for standard clinical interoperability formats and for batch JSONL is provided with the reproducibility artifact. Specific transport-layer choices are deployment-site decisions and are not part of the framework.

5.3 Latency and failure modes

The per-patient update path was profiled on a 10 000-patient stream with 13 marginal cells. The measured latency was below 10 ms on a single CPU core. Ensemble decomposition and dashboard rendering were excluded from the profile. Under these experimental settings, the core e-process monitor is unlikely to dominate the computational cost. Three failure modes are handled explicitly. A missing protected attribute drops the affected record from the intersectional monitor while the marginal monitor continues. An ensemble crash skips the decomposition while the e-process continues. A corrupted counterfactual records the indicator as missing and extends the warm-up window.

5.4 PHI handling and configuration

The framework receives de-identified records by contract and does not store raw protected health information. Audit logs are append-only and contain only e-values, cell labels, and timestamps. Default operating points are $\alpha = 0.05$ for the anytime-valid threshold, $q = 0.10$ for the inspection-time Benjamini-Hochberg level, $n_{\text{warm}} = 100$ for the e-process warm-up, $c = 0.25$ for the GROW shrinkage, $K = 11$ ensemble members, and $\tau^\star = 5\,000$ patients for the RMFT horizon. Further implementation details, configuration files, random seeds, and scripts used to reproduce the reported experiments will be provided in the reproducibility artifact, subject to anonymisation, institutional approval, and journal artifact-release requirements.

5.5 Deployment overhead and operational cost

Computational overhead. The per-patient e-process update is $O(1)$ per monitored cell. The inspection-time step-wise Benjamini-Hochberg sort across m cells is $O(m \log m)$. Flip Hazard is updated incrementally on each arrival or refreshed periodically. Diagnostic attribution requires ensemble predictions and is more expensive per patient than the e-process. RCAP is batch or on-demand because it operates on group-conditional rank-position distributions. Dashboard rendering is decoupled from the core monitoring path.

Table 3. Operational comparison of post-release monitoring options. Axes are operational rather than statistical, and the diagnostic-output column is restricted to triage signals rather than confirmed causal attribution.

Method	Monitoring cadence	Core statistic	Per-patient update	Repeated-inspection control	Diagnostic output	Operational
Static VFR	pre-release or periodic batch	resample-flip fraction	none during stream	no	no	no time-loca
Periodic batch test	fixed batches	two-proportion test on each batch	$O(\textit{batch size})$ at batch boundaries	no, unless separately corrected	no	peeking ar latency trade
ADWIN	streaming	windowed Hoeffding bound	lightweight per patient	drift-detector bound; not fairness-specific anytime-valid verdict evidence	no	not designed terfactual verdict flips
CHRONO-Fair	streaming plus on-demand modules	per-cell shrunk-GROW e-process; Flip Hazard; decomposition; RCAP	$O(1)$ per cell for e-process; $O(m \log m)$ for inspection-time BH	per-cell anytime-valid Type-I control	yes, triage signal only	controlled r dence; prosp idation need

Runtime. The per-patient update path was profiled on a 10 000-patient stream with 13 marginal cells. The measured latency was below 10 ms on a single CPU core. Ensemble decomposition and dashboard rendering were excluded from the profile.

Storage overhead. A minimal audit record stores the arrival timestamp, model version, protected-cell label, factual prediction, counterfactual prediction, flip indicator, current e-value, and alarm status. Diagnostic-attribution summaries are stored optionally. No raw protected health information is required by the monitoring layer if prediction outputs and protected-attribute audit labels are provided under governance controls. Append-only audit logs are sufficient for reproducibility.

Human labour overhead. CHRONO-Fair does not remove governance review. It reduces the manual inspection burden by summarising, for each flagged cell, the baseline flip rate, the current flip rate, the e-value, the diagnostic-attribution signal, a recommended review action, and explicit caveats.

Operational comparison with baselines. Table 3 compares CHRONO-Fair with the three baseline monitoring options used elsewhere in the paper, on operational axes that affect deployment cost. CHRONO-Fair is computationally lightweight for the core e-process. Diagnostic attribution and RCAP are optional, on-demand modules that add interpretive value at additional cost.

5.6 Artifact availability and reproducibility

CHRONO-Fair is accompanied by a proof-of-concept research prototype intended to support reproducibility and inspection of the monitoring workflow. The prototype implements the stream interface, the counterfactual flip indicator, the per-cell e-process update, the Flip Hazard estimator, the diagnostic-attribution module, the RCAP statistic, and the Inspector report generation. It is not intended as an operational clinical system or medical device. Implementation details, including the Python environment, package dependencies, configuration files, random seeds, and scripts used to reproduce the reported experiments, will be provided in the artifact documentation rather than treated as methodological contributions.

5.7 Scope statement

The deployment description above specifies the contract of a research prototype, not the certification of a production system. Hospital integration would require institutional review-board approval, local validation on the deployment site's population, and human-in-the-loop review of every alarm. The framework is not a medical device.

6 EXPERIMENTS

The evidence base of this study is organised as three layers that decline in directness from real clinical data toward controlled simulation. Layer 1 is the real Texas-100X batch verdict-stability analysis of Section 6.8. Layer 2 is the set of Texas-100X-calibrated controlled streaming and retrospective replay experiments. These characterise sequential monitoring behaviour under known drift settings. They include the quarter-based and hospital-group retrospective replays of Section 6.16. The replays approximate deployment ordering on the real cohort but do not replace live clinical validation. Layer 3 is a robustness and stress-test layer covering baseline miscalibration, small intersectional cells, and label delay. Only Layer 1 uses real clinical records directly; Layers 2 and 3 evaluate sequential behaviour under controlled replay or simulation. Texas-100X is administrative discharge data rather than a live admission stream, so prospective EHR-stream validation remains future work.

What the framework is compared against, and where each comparator comes from. The post-deployment monitoring comparison is run on a single shared substrate: the same Texas-100X-calibrated stream, the same Fair monitored model (Section 4.6), the same patient-level counterfactual flip indicator. Holding the substrate fixed means the reported gap between methods reflects the alarm rule, not the data path. Five comparator rules are drawn from named published instruments, and each enters the comparison as published rather than re-derived.

The snapshot reference is the scalar Verdict Flip Rate of the prior work [7], computed once on a pre-deployment validation window. It is the simplest possible monitor: a single number, never refreshed after release. Its purpose is to show how much information a deployment loses by not updating at all. Sitting one step above the snapshot is the quarterly-audit instrument of Davis et al. [29], here reproduced as a two-proportion z-test on non-overlapping batches of 500 patients. The Davis paper ran its test at a calendar-quarter cadence on a VA cohort; the same statistical object is used here, with the cadence converted to a patient count to fit a single-site deployment. The change-point family is represented by ADWIN of Bifet and Gavalda [12], run on the per-patient flip indicator at two inspection schedules to separate cadence effects from rule effects. A fairness-gap variant of ADWIN, in the spirit of the JAMIA 2025 fairness-drift recommendation [29], applies the same detector to the centred flip series rather than to the raw indicator. The closest direct precedent on the counterfactual side is Prediction Sensitivity [75], which reports a per-batch counterfactual-flip rate without an alarm rule; it appears in the comparison as a methods-level reference, since the published version is an audit rather than a sequential alarm.

The framework introduced in this paper sits beside those rules rather than replacing them. Its statistical content is the per-cell shrunk-GROW e-process under Ville's inequality with a step-wise Benjamini-Hochberg cross-cell layer at each inspection time, plus the diagnostic aleatoric-epistemic split and the RCAP rank-allocation statistic. Holding the deployed model and the flip indicator constant across all six rules, the post-deployment comparison in Section 6.2 therefore reads as a head-to-head of alarm rules under matched conditions.

The Fair monitored model used in the streaming experiments is described in Section 6.10, where the Standard-versus-Fair monitored-model setup is tabulated. The Standard-to-Fair transition is taken from the prior VFR-Audit study [7] as the deployed-model configuration that CHRONO-Fair

Experiment	n	Trials	Drift setting	Metric	Baseline(s)
Exp. 1 detection delay	10 000	50	onset at $t = 5\,000$; $\rho_0 = 0.05 \to \rho_1 = 0.15$	patients-to-alarm; detection rate	ADWIN (k=1, k=25), Davis-ADWIN, periodic batch (500), static VFR
Exp. 2 decomposition	1 000	30/scenario	3 scenarios (aleatoric, epistemic, mixed)	confusion-matrix accuracy	none (validation against ground truth)
Exp. 3 Flip Hazard	15 000	1	no drift, structural disparity	RMFT, log-rank χ^2, KM survival	scalar VFR [7]
Exp. 4 RCAP	20 000	1 (1 000 bootstraps)	no drift, structural disparity	Wasserstein-1 rank shift, MAE gap	group-MAE gap
Exp. 5 false-alarm	10 000	200/α	no drift (H_0)	empirical false-alarm rate per nominal α	nominal α line
Exp. 6 sensitivity	12 000	30/ρ_1	onset at $t = 6\,000$; $\rho_1 \in [0.06, 0.20]$	detection rate; mean delay vs drift size	none (characterisation)
Exp. 7 four-attribute audit	15 000	1	no drift; structural racial disparity	log-rank χ^2 across race, ethnicity, sex, age	marginal flip rate
Exp. 8 real Texas-100X VFR	925 128	1000 boot.	real audit (no simulation)	point vs CI vs VFR verdict stability	point estimate; bootstrap CI
Exp. 9 robustness stress tests	200 to 12 000	100/setting	miscalibration, small cell, label delay	false-alarm rate; detection rate; delay	correctly-calibrated monitor
Exp. 10 multi-model fairness	925 128	1	real test set; LR, RF, GB, XGBoost, LightGBM, DNN	DI, WTPR, SPD, EOD, PPV across 4 attrs	six models
Exp. 11 temporal/hospital replay	30 000; 67 330	1	quarter drift; hospital case-mix; real record order	alarm; detection delay	ADWIN; static VFR
Exp. 12 Standard/Fair monitored-model setup	925 128	1	defines the deployed Fair model that CHRONO-Fair monitors	DI before and after pre-release static audit, across 4 attributes	Standard (uncorrected) model

Table 4. Experiment summary. Each row lists stream length, replication count, drift configuration, primary metric, and baselines.

monitors. A cross-study numerical comparison against other published length-of-stay fairness work was not undertaken. No prior length-of-stay study runs a per-cell anytime-valid monitor or a Flip Hazard estimator on the same substrate. A head-to-head number across studies would therefore be meaningless.

Table 4 summarises each experiment.

6.1 Synthesiser

A synthetic stream calibrated to the Texas-100X record statistics [54] was generated. The mix matched the dataset's published demographics: White 48%, Hispanic 30%, Black 13%, Asian/PI 5%, Other 4%; Female 55%, Male 45%; Non-Hispanic 65%, Hispanic 35%; age groups Pediatric 5%, Young Adult 20%, Middle-aged 40%, Elderly 35%. Continuous LOS was generated as $\text{LOS} = 4.2 + \beta^\top X + \epsilon$ with $\epsilon \sim \mathcal{N}(0, 0.5^2)$. The binary target $y_{\text{ext}} = \mathbf{1}\{\text{LOS} > 3\}$ matched the prior implementation's class balance of 55% normal stay, 45% extended stay. A structural disparity of +0.4 on the first three risk features was injected for Black and Hispanic groups, reproducing the bias structure documented in the prior VFR-Audit work. Optional knobs controlled drift onset and aleatoric label noise.

6.2 Detection delay

Setup. A stream of $n = 10\,000$ patients was generated under $\rho_0 = 0.05$ until index $t = 5\,000$ and under $\rho_1 = 0.15$ thereafter. The CHRONO-Fair e-process was run alongside five baselines. ADWIN with check-every-patient ($k = 1$) and ADWIN with check-every-25-patients ($k = 25$) provide two ADWIN cadences. A Davis-style ADWIN on the binary fairness gap series [29] provides a fairness-gap variant. A periodic two-proportion z-test on non-overlapping batches of 500 patients provides a batch baseline. A static VFR snapshot computed once before deployment provides the no-update baseline. Fifty Monte-Carlo replications were run with seeds 0 to 49.

Metric. For each method and replication, the number of patients between drift onset at $t = 5\,000$ and the first alarm was recorded. Runs that produced an alarm before drift onset were counted as false alarms and excluded from the delay calculation.

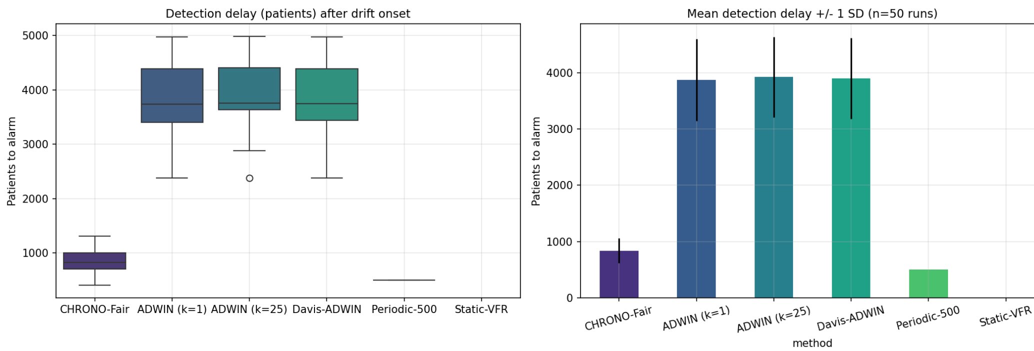

Fig. 3. Section 6.2. Detection delay across 50 Monte-Carlo runs at $n = 10\,000$, drift onset at $t = 5\,000$. CHRONO-Fair: 50 of 50 alarms at mean delay 837. ADWIN variants and Davis-ADWIN: 21 of 50 at mean delay 3 878 to 3 927. Periodic batch (500): 24 of 50 at mean 499 without peeking control. Static VFR: 0 of 50.

Result. CHRONO-Fair alarmed in 50 of 50 replications with a mean delay of 837 patients (median 828, standard deviation 221). ADWIN at $k = 1$ alarmed in 21 of 50 replications at mean delay 3 878 patients. ADWIN at $k = 25$ alarmed in 21 of 50 at mean 3 927. Davis-ADWIN alarmed in 21 of 50 at mean 3 904. The periodic batch test alarmed in 24 of 50 at mean 499 patients but did not control false alarms across repeated peeks. Static VFR did not alarm. Under a separate no-disparity sanity scenario at $n = 10\,000$ and seeds 10 000 to 10 049, CHRONO-Fair raised 0 alarms in 50 replications.

Interpretation. The detection-rate gap (50 of 50 against 21 of 50) is larger than the delay gap and is the headline result. ADWIN's coverage at $k = 1$ and $k = 25$ is similar, indicating that the framework's advantage is not an artefact of check-frequency. The periodic batch test trades coverage for delay and provides weaker support for the continuous-monitoring posture motivated by FDA PCCP guidance [104].

Figure 3 plots the per-method detection delay across the 50 runs.

6.3 Decomposition accuracy

Setup. Two tracks were run. Track 1 generated ensembles of 15 binary classifiers whose per-instance probabilities were constructed by design to be aleatoric-dominated, epistemic-dominated, or mixed. Thirty replications per scenario were generated with seeds 0 to 29. Track 2 trained ensembles of 11 single decision trees with depth 6, each on a disjoint slice of the minority training data (epistemic by construction) or the full training data with 0.25 aleatoric label flip (aleatoric by construction). Fifteen replications per scenario were run.

Metric. The Inspector Agent label (one of `aleatoric`, `epistemic`, `mixed`, `none`) was compared against the ground-truth scenario. The aggregate confusion matrix and accuracy were reported.

Result. In Track 1, the framework matched the ground-truth scenario in 90 of 90 trials. In Track 2, the ML-driven setup, the framework labelled the dominant diagnostic signal as `aleatoric` in 3 of 15 aleatoric trials, 1 of 15 epistemic trials, and 1 of 15 mixed trials; the remaining trials were labelled `mixed`. Net Track 2 accuracy was 37.8%.

Interpretation. The decomposition formula is correct on inputs that satisfy the controlled-ensemble assumption (Track 1). In realistic ensembles trained on overlapping data and audited at deployment time, the dominant component is frequently `mixed`. The framework's recommended action under `mixed` is parallel data-pipeline audit and ensemble-stabilised retraining, which is the same action a clinical safety officer would take when the diagnostic interpretation is unclear. The

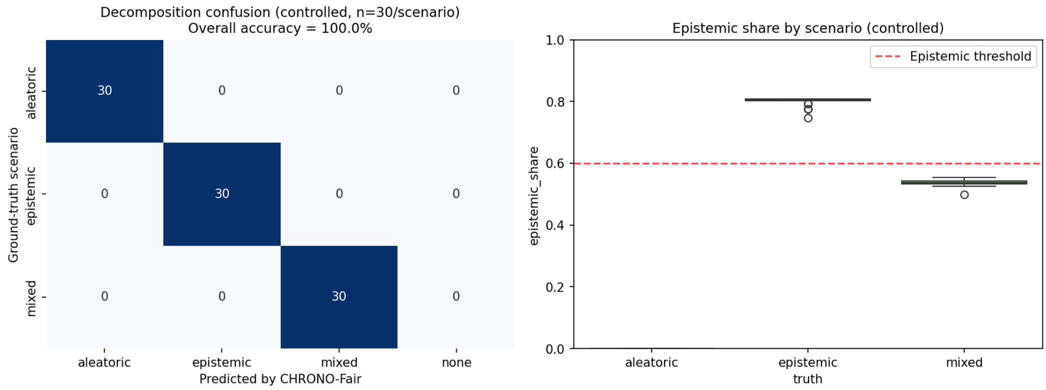

Fig. 4. Section 6.3 Track 1. Controlled decomposition validation, 30 trials per scenario at $n = 1\,000$. Confusion matrix on the left; per-scenario epistemic share on the right. The 0.6 threshold is the Inspector Agent decision boundary.

Track 2 figure should be read not as an accuracy bound but as evidence that the realistic case is mixed.

Figure 4 plots the controlled-validation confusion matrix and the per-scenario epistemic share.

6.4 Flip Hazard survival on Texas-100X-style data

Setup. A stream of $n = 15\,000$ patients was generated with seed 42. The deployed classifier was a logistic regression trained on five risk features. Counterfactual predictions were generated by removing the +0.4 minority shift on features 0 to 2. The Flip Hazard estimator was run per protected group with a horizon $\tau^\star = 5\,000$.

Metric. Per-group Kaplan-Meier no-flip survival curves with Greenwood bands, RMFT at $\tau^\star = 5\,000$, and pairwise log-rank statistics against the White reference group were reported.

Result. RMFT was 4 817 patients for White ($n = 7\,214$), 3 940 for Other ($n = 588$), 3 837 for Hispanic ($n = 4\,518$), 3 337 for Asian/PI ($n = 734$), and 2 459 for Black ($n = 1\,946$). The scalar VFR was 0.078 for White, 0.066 for Other, 0.201 for Hispanic, 0.065 for Asian/PI, 0.190 for Black. Log-rank statistics against White were $\chi^2 = 1\,675.49$ (Black), $\chi^2 = 1\,248.21$ (Hispanic), $\chi^2 = 456.33$ (Asian/PI), $\chi^2 = 455.75$ (Other), all with $p < 10^{-10}$.

Interpretation. The scalar VFR collapses the Black-Hispanic ordering (0.190 against 0.201). The RMFT preserves the ordering: Black has the lowest RMFT (2 459), then Asian/PI (3 337), then Hispanic (3 837), then Other (3 940), then White (4 817). RMFT distinguishes early-clustering disparity (Black: a flip cluster appears within the first 2 500 arrivals) from late-clustering disparity (Hispanic: flips spread over a longer arrival window). The clinical meaning is the same in both cases (disparate exposure), but the operational response differs: an early cluster triggers an immediate review, a late cluster triggers a slower investigation.

Figures 5 to 7 plot the intersectional curves, the group-stratified survival curves, and the RMFT comparison against scalar VFR.

6.5 RCAP for length-of-stay regression

Setup. A stream of $n = 20\,000$ patients was generated with seed 11. A Ridge regressor (regularisation $\alpha = 1.0$) was trained on the first 8 000 records to predict LOS. The remaining 12 000 were used for audit. RCAP was computed pairwise across the five racial groups with $B = 1\,000$ bootstrap replicates

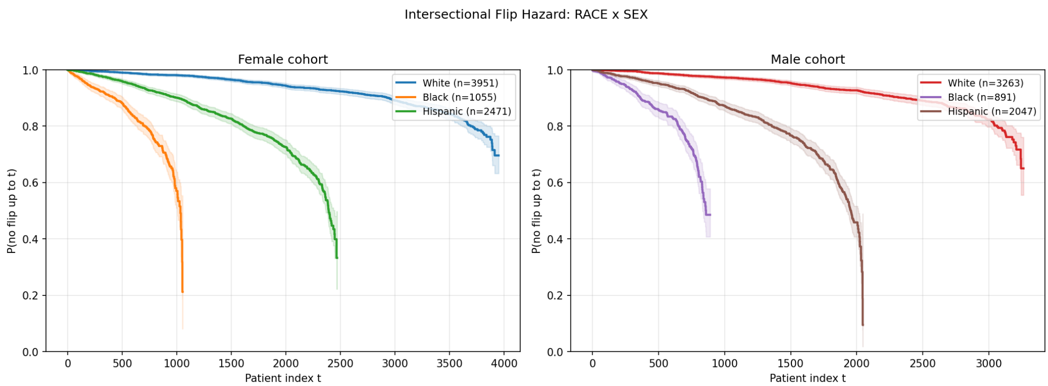

Fig. 5. Section 6.4. Intersectional Flip Hazard: race by sex.

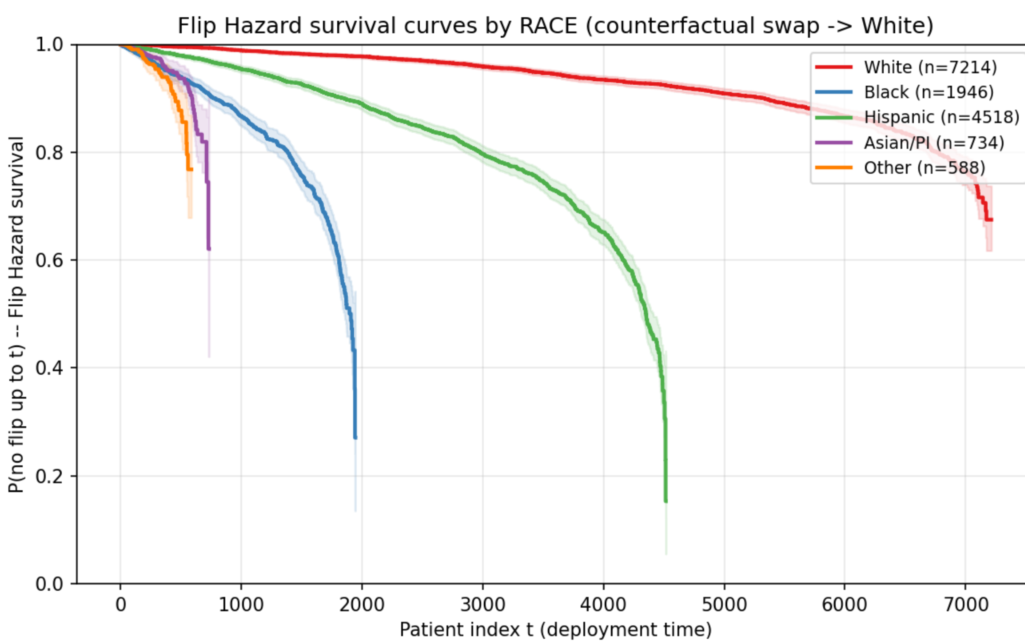

Fig. 6. Section 6.4. Group-stratified Flip Hazard no-flip survival at $n = 15\,000$. Greenwood 95% bands shown.

for the confidence interval. The group-MAE gap against the White reference was computed as a baseline.

Metric. Wasserstein-1 distance between counterfactual rank distributions per group pair, MAE gap against White, and per-patient rank shift histograms.

Result. RCAP-against-White was computed as the Wasserstein-1 distance between the factual and counterfactual rank-position distributions, with 1 000 bootstrap resamples per group. The empirical Wasserstein-1 plug-in is biased upward at small sample size, so the plug-in estimate, the bootstrap mean, and the percentile interval are reported separately rather than as a symmetric interval around the plug-in. Table 5 gives the four values per group. For Black ($n = 1\,488$) the

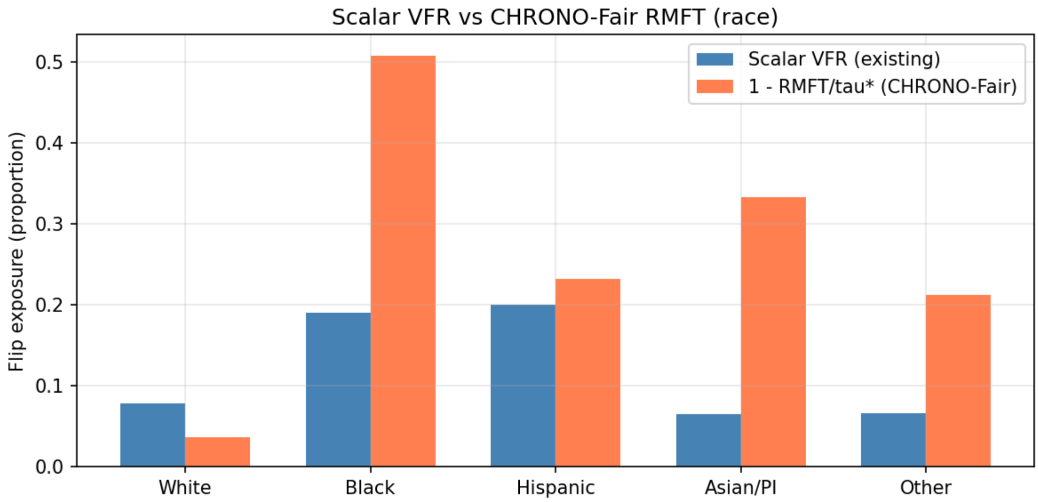

Fig. 7. Section 6.4. Scalar VFR (left bars) against $1 - \text{RMFT}/\tau^{\star}$ (right bars). RMFT preserves the Black versus Hispanic ordering that the scalar VFR collapses.

Table 5. RCAP against the White reference. The plug-in is the Wasserstein-1 distance on the observed rank positions; the bootstrap mean and the 95% percentile interval come from 1 000 resamples. The bootstrap mean exceeds the plug-in for every group, which quantifies the upward small-sample bias of the empirical Wasserstein-1 estimator.

Group	n	Plug-in W_1	Boot. mean	95% interval
Black	1 488	0.0163	0.0197	[0.0088, 0.0336]
Asian/PI	552	0.0142	0.0233	[0.0117, 0.0425]
Hispanic	3 643	0.0066	0.0099	[0.0052, 0.0169]
Other	499	0.0049	0.0208	[0.0100, 0.0429]

plug-in W_1 was 0.0163, the bootstrap mean was 0.0197, and the 95% percentile interval was [0.0088, 0.0336]. For Other ($n = 499$) the plug-in was 0.0049, the bootstrap mean was 0.0208, and the interval was [0.0100, 0.0429]; the percentile interval lies above the plug-in because the upward bias of the empirical estimator dominates at $n = 499$. The MAE gap against White was 0.0065 days for Black, 0.0062 for Asian/PI, 0.0006 for Hispanic, and 0.0028 for Other. The MAE ordering was Black, Asian/PI, Other, Hispanic, while the plug-in RCAP ordering was Black, Asian/PI, Hispanic, Other; the two orderings agreed on the leader, Black, but disagreed on the placement of Hispanic and Other.

Interpretation. The MAE gap measures average error, which a regression model can keep low even while systematically shifting one group's allocation rank. RCAP captures the rank shift directly. The clinical reading is that, under the counterfactual swap, the Black rank-position distribution shifts by 1.63 percentile points from the White-referenced distribution on the plug-in estimate, with a 95% percentile interval of 0.88 to 3.36 percentile points that excludes zero. The triage and bed-allocation literature treats predicted LOS as an ordinal allocation signal, so a rank-position shift is the policy-relevant quantity [102]. Because the empirical Wasserstein-1 estimator is upward-biased

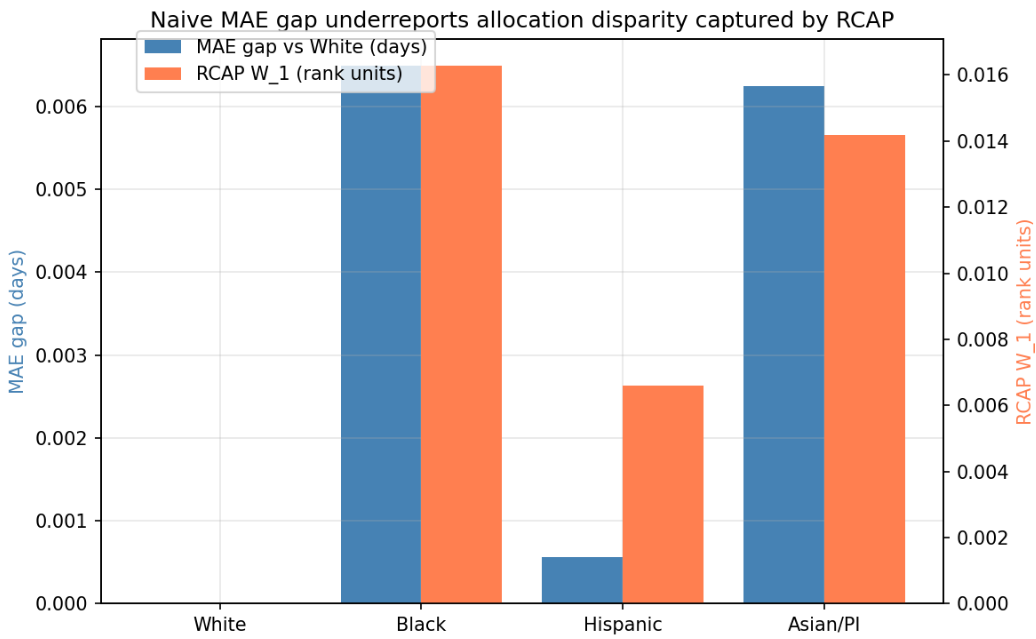

Fig. 8. Section 6.5. Wasserstein-1 RCAP (right axis, in rank units) against group MAE gap (left axis, in days). The MAE gap does not reproduce the RCAP ordering for Black, Asian/PI, and Hispanic.

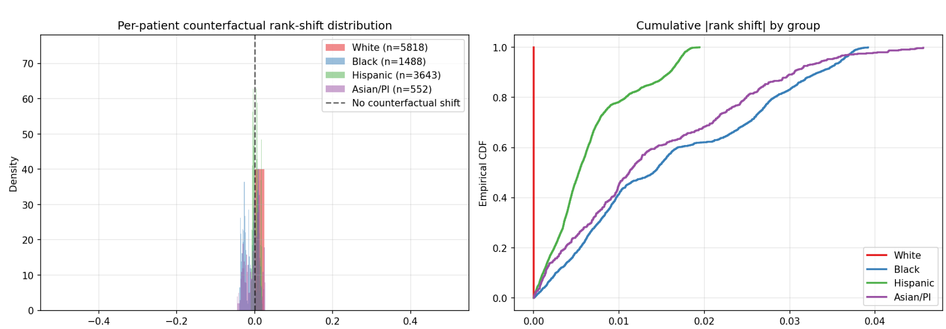

Fig. 9. Section 6.5. RCAP per-patient counterfactual rank shifts at $n = 20\,000$. Left: density. Right: cumulative $|\Delta|$.

at small n, RCAP must be read together with the per-cell sample size, and cells near $n = 500$ such as Other should be merged or flagged as low-power rather than interpreted at face value.

Figures 8 and 9 plot the RCAP-versus-MAE-gap comparison and the per-patient rank-shift distribution.

6.6 False-alarm calibration

Setup. Under the no-drift null H_0, 200 independent streams of length $n = 10\,000$ were generated for each nominal $\alpha \in \{0.01, 0.025, 0.05, 0.10, 0.20\}$. Seeds were 0 to 199 for each level. CHRONO-Fair was run with shrinkage $c = 0.25$ and warm-up $n_{\text{warm}} = 100$.

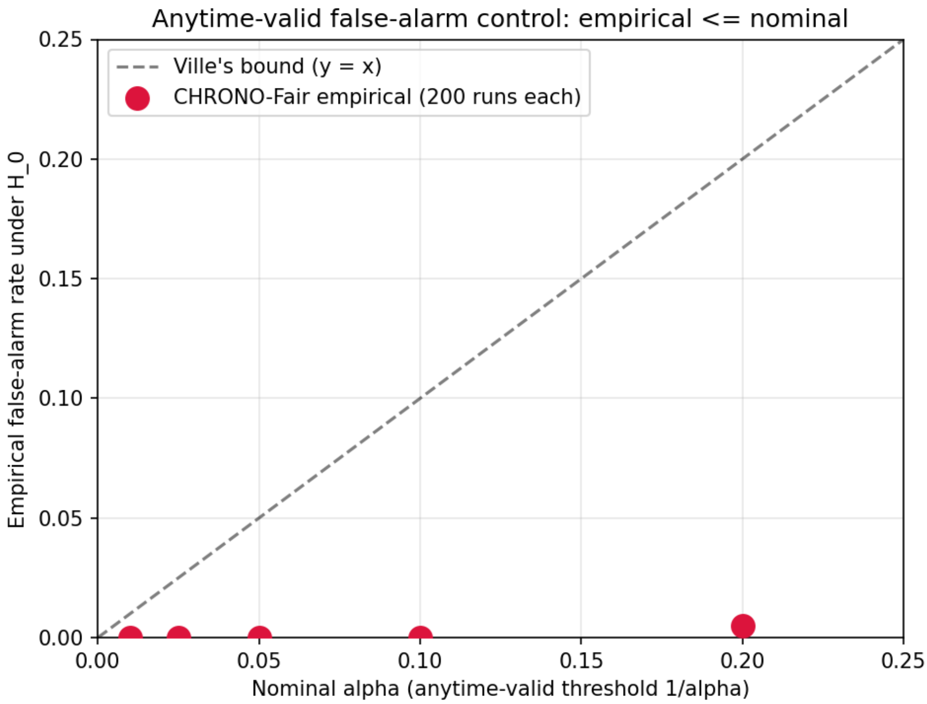

Fig. 10. Section 6.6. Empirical false-alarm rate against nominal α on 200 streams per level. Points lie on or below the $y = x$ Ville bound.

Metric. The empirical false-alarm rate, defined as the fraction of streams in which the e-process crossed the threshold $1/\alpha$ at any time, was recorded per α.

Result. The empirical false-alarm rates were 0.000 ($\alpha = 0.01$), 0.000 ($\alpha = 0.025$), 0.000 ($\alpha = 0.05$), 0.000 ($\alpha = 0.10$), and 0.005 ($\alpha = 0.20$). All values were at or below the nominal α. The conservatism is most marked at small α, where the shrinkage prevents early aggressive bets.

Interpretation. The empirical-vs-nominal alignment confirms Theorem 4.2 on the synthesiser. Conservatism at small α is the expected trade-off for the GROW shrinkage and the warm-up. Detection-delay results in Section 6.2 were obtained at $\alpha = 0.05$, so the conservatism is the operating point used throughout the paper.

Figure 10 plots the empirical false-alarm rate against the nominal level for each tested α.

6.7 Live dashboard rendering

Setup. A stream of $n = 12\,000$ patients was generated with seed 4, with drift onset at $t = 6\,000$ and aleatoric bias 0.06. Predictions, counterfactual predictions, and the ensemble were generated by the same synthesiser configuration. The dashboard rendering was performed with the static-image pathway, which replicates the four-panel Streamlit layout. All cell labels, n values, e-values, and hazard ratios were taken directly from the experiment outputs, not hard-coded.

Result. The dashboard exposes four panels populated from the live experiment state. Panel 1 shows Flip Hazard survival per group. Panel 2 shows the e-process log E_t trace per group with the $\log(1/\alpha)$ alarm line. Panel 3 shows the aleatoric/epistemic split per group as stacked bars. Panel 4 shows the RCAP per-patient rank-shift histogram per group. The Inspector Agent action card lists

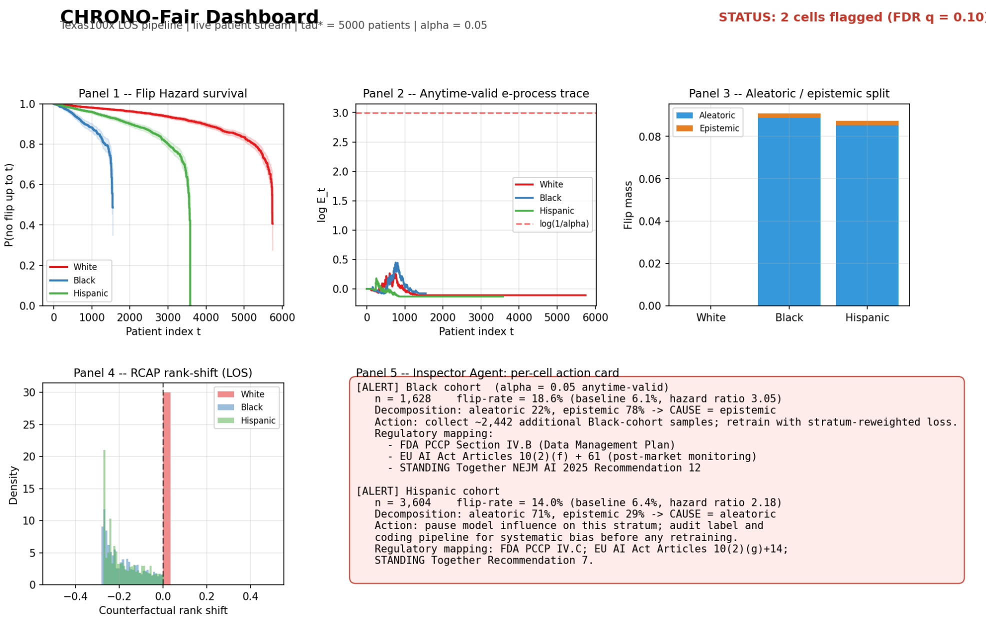

Fig. 11. Section 6.7. Static rendering of the four-panel dashboard at $n = 12\,000$ with drift onset at $t = 6\,000$. Inspector Agent card lists cell labels, e-values, hazard ratios, dominant triage signal, recommended action, and suggested regulatory mapping.

the flagged cells with their diagnostic triage label, recommended action, and suggested regulatory mapping.

Interpretation. The dashboard prototype is designed to support the type of inspectable monitoring output expected in FDA PCCP-style lifecycle review [104], namely that monitoring outputs be inspectable by a clinical governance team without machine-learning expertise. Replication on a hospital information system uses the data interface described in Section 5.

Figure 11 shows the four-panel dashboard rendered on this stream.

6.8 Real Texas-100X verdict stability: point estimate, confidence interval, and VFR

Setup. This experiment uses real fairness results, not synthetic streams. The prior VFR-Audit implementation trained four classifiers (logistic regression, random forest, gradient boosting, a two-layer neural network) on the full Texas-100X discharge dataset. The dataset has 925 128 records and a 185 026-record held-out test set. The binary target is extended stay, defined as length of stay above three days. Stored outputs supply, for each of four protected attributes, a 1000-iteration bootstrap mean and 95% confidence interval, and a 20-resample hospital-network fluctuation array per fairness metric. Three verdict procedures were compared on the same stored numbers. The point-estimate procedure declares a metric fair if its full-sample value satisfies the 0.8 rule. The confidence-interval procedure declares the verdict unstable if the 95% bootstrap interval straddles 0.8. The Verdict Flip Rate is the fraction of the 20 resamples whose verdict differs from the resample majority. The 0.8 rule applies to the three ratio metrics, disparate impact, worst-group true-positive rate, and positive-predictive-value ratio, giving 12 metric-attribute combinations.

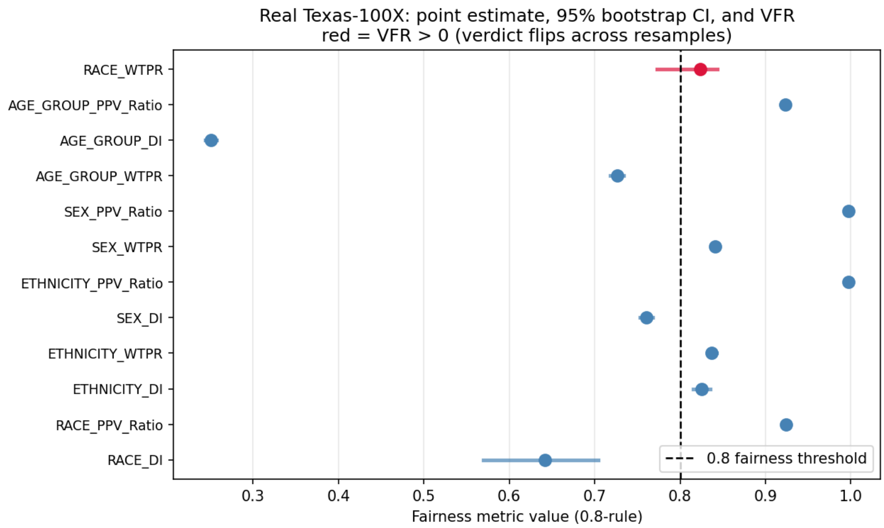

Fig. 12. Section 6.8. Real Texas-100X verdict stability. Point estimate (marker) and 95% bootstrap confidence interval (bar) for 12 metric-attribute combinations, against the 0.8 fairness line. Red marks the one combination with a non-zero Verdict Flip Rate, race worst-group true-positive rate, where the CI straddles 0.8 and 4 of 20 resamples flip the verdict.

Metric. Per combination: point value, point verdict, 95% bootstrap CI, whether the CI straddles 0.8, the resample fair-unfair split, and the VFR. Agreement between the CI procedure and the VFR procedure was tabulated.

Result. Of the 12 audited combinations, 11 had a stable verdict. Their bootstrap CIs lay entirely on one side of 0.8 and their VFR was 0.000. One combination was unstable. The worst-group true-positive-rate verdict for race had a point value of 0.824, which the point-estimate procedure read as fair. Its 95% bootstrap CI was 0.773 to 0.844, straddling 0.8. Its 20-resample split was 16 fair and 4 unfair, a VFR of 0.200. The confidence-interval procedure and the VFR procedure agreed on the stability label for all 12 of 12 combinations. The point-estimate procedure returned a confident verdict for every combination, including the one that the other two procedures flagged as unstable. Figure 12 shows the point estimate, the 95% CI, and the 0.8 line for all 12 combinations.

Interpretation. The result establishes three points on real data. First, verdict instability is real and not a simulation artefact. One in 12 audited verdicts on Texas-100X flips across hospital-network resamples. Second, the confidence interval and the Verdict Flip Rate are consistent. They agreed on every combination, which supports using either as a batch instability test. Third, the point estimate alone is unsafe. It reported the race worst-group true-positive-rate verdict as fair, hiding a 1-in-5 resample flip. This is the case for sequential monitoring. A confidence interval or a VFR is computed once before deployment. Neither says whether, on a live patient stream after deployment, the same verdict has crossed 0.8 again, nor when, nor why. The CHRONO-Fair Flip Hazard estimator and e-process (Sections 6.4, 6.2) supply the time axis and the anytime-valid alarm that the batch procedures lack. The decomposition (Section 6.3) supplies the diagnostic triage.

6.9 Multi-model, multi-attribute fairness on real Texas-100X

Evaluation question. Does the verdict instability shown in Section 6.8 reflect one fragile combination of model and attribute, or a pattern that repeats across model families and protected attributes?

Table 6. Multi-model multi-attribute fairness on real Texas-100X. Each cell is the count of fair metrics out of six (DI, WTPR, SPD, EOD, PPV_Ratio, Eq_Odds) at the 0.8 ratio rule and the 0.1 gap rule. Higher is fairer. Numbers are derived from the stored real test-set fairness summary.

Model	RACE	ETHNICITY	SEX	AGE_GROUP
LR	1	2	2	1
RF	2	6	3	1
GB	2	6	4	1
XGBoost	4	6	4	1
LightGBM	4	6	4	1
DNN	2	6	3	1

CHRONO-Fair monitors the thresholded verdict; this subsection examines what the underlying verdicts look like across all four protected attributes for six modelling choices.

Setup. Stored fairness numbers from the prior VFR-Audit implementation are used. Six model variants trained on the full Texas-100X training set and evaluated on the 185 026-record held-out test set are compared: logistic regression (LR), random forest (RF), gradient boosting (GB), XGBoost, LightGBM, and a PyTorch deep neural network (DNN). For each model, six fairness metrics are computed across the four protected attributes RACE, ETHNICITY, SEX, and AGE_GROUP: disparate impact (DI), worst-group true-positive rate (WTPR), statistical-parity difference (SPD), equalised-odds difference (EOD), positive-predictive-value ratio (PPV_Ratio), and the aggregate equalised-odds value (Eq_Odds).

Metric. The fairness verdict for each (model, attribute, metric) cell. The 0.8 rule is applied to DI, WTPR, and PPV_Ratio. A 0.1 gap rule is applied to SPD, EOD, and Eq_Odds. The cell summary in Table 6 reports the count of fair metrics out of six for each (model, attribute) cell.

Result. ETHNICITY is the easiest protected attribute: every tree-based or DNN model passes six of six metrics, and LR passes two. AGE_GROUP is uniformly the hardest: every model passes exactly one metric, the positive-predictive-value ratio, and fails the other five. RACE divides the models clearly: XGBoost and LightGBM pass four of six metrics, the tree-based RF, GB, and the DNN pass two, and LR passes only one. SEX falls between, with XGBoost, GB, and LightGBM passing four of six, and LR passing two. Figure 13 shows the underlying DI, WTPR, SPD, and EOD values across the same grid.

Interpretation. Two findings have direct bearing on the monitoring framework. First, the per-attribute fairness profile is not a model-family invariant. On RACE and SEX, XGBoost and LightGBM each pass four of six metrics, while LR passes one and two respectively. On AGE_GROUP, every model passes one of six metrics. The persistent AGE_GROUP failure across six model families points to a data-side rather than a model-side signal. Second, the race worst-group true-positive-rate verdict that flips across resamples in Section 6.8 is itself a cell in this grid. XGBoost RACE WTPR passes here, LR RACE WTPR fails. The framework's diagnostic decomposition is intended to flag whether such a swap reflects label bias on the affected attribute (an aleatoric signal) or limited sample size (an epistemic signal). The grid therefore motivates per-cell monitoring across all four protected attributes simultaneously rather than racial-cell monitoring alone. This is the default configuration of the Live Stream tab in the released dashboard.

6.10 Monitored-model setup: Standard versus Fair

Setup. The streaming experiments require a deployed model whose post-release fairness can be monitored. Two model states are defined for that purpose. The Standard model is the uncorrected

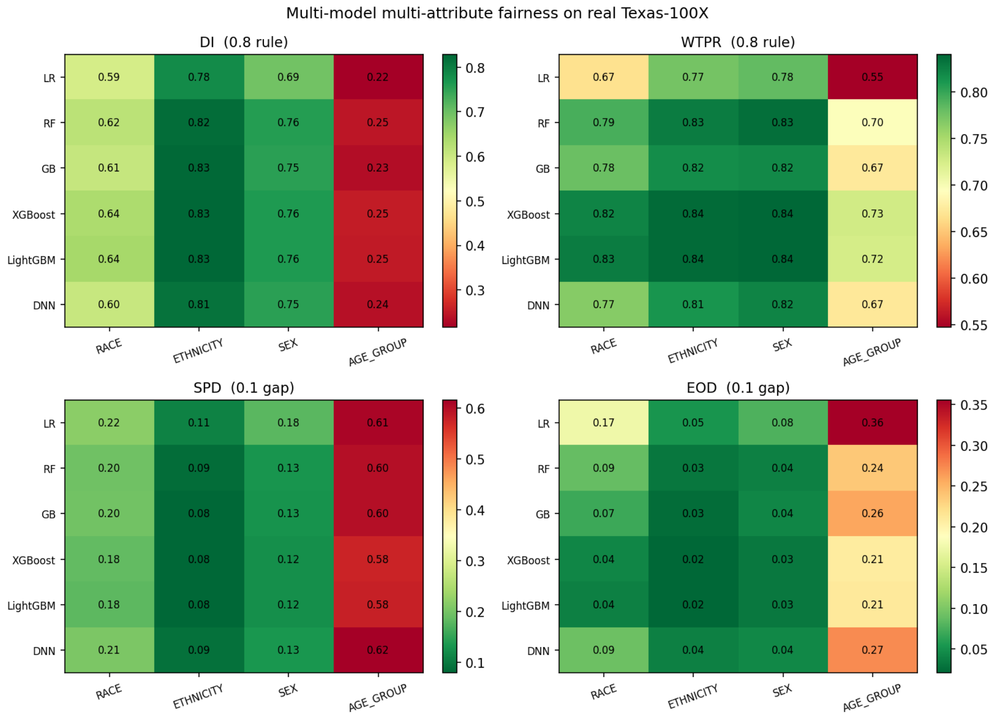

Fig. 13. Section 6.9. Multi-model multi-attribute fairness on real Texas-100X. Each panel is one metric (DI, WTPR, SPD, EOD) across six models and four protected attributes. Cell value is the metric; colour encodes pass-or-fail under the 0.8 rule for the ratio metrics and the 0.1 gap rule for SPD/EOD.

pre-release model. The Fair monitored model is the deployed configuration adopted after the static VFR-Audit-style pre-release fairness audit reported in [7]. CHRONO-Fair does not introduce this correction procedure. It monitors whether the adopted Fair verdict remains stable after deployment.

Metric. Disparate impact (DI) is computed per protected attribute on the real Texas-100X test set. A cell passes the four-fifths rule when DI $\geq$ 0.80. Overall accuracy is reported alongside DI for both the Standard and the Fair monitored model.

Result. Table 7 reports the Standard-versus-Fair monitored-model setup. The Standard model fails the DI rule on Race, Sex, and Age Group, and passes only on Ethnicity. The Fair monitored model satisfies DI $\geq$ 0.80 on all four protected attributes. The Fair monitored model reduces overall accuracy by 5.28 percentage points relative to the Standard model. These values are taken directly from the prior VFR-Audit study [7] and are reproduced here as the deployed-model configuration that CHRONO-Fair monitors. They are not a CHRONO-Fair contribution.

Interpretation. The point is not that CHRONO-Fair produces the Fair model. The point is that even after a model is released with passing point-estimate fairness verdicts on all four protected attributes, those verdicts may later become unstable under new patients, hospital case-mix changes, or subgroup drift. CHRONO-Fair monitors that post-release risk. Age Group is the binding protected attribute in the Standard model. It remains the highest-priority cell for post-release monitoring even after the Fair model passes the point-estimate DI threshold. The largest correction movement is the cell most likely to drift back under deployment conditions.

Table 7. Pre-release monitored-model setup. The Standard model represents the uncorrected model before fairness adjustment. The Fair monitored model is the deployed configuration monitored by CHRONO-Fair. The correction step is not a contribution of CHRONO-Fair; it defines the starting condition for post-deployment monitoring.

Attribute	Std DI	Fair DI	Δ DI	Std verdict	Fair verdict	Source
Race	0.654	0.884	+0.230	fail	pass	prior VFR-Audit setup
Sex	0.762	0.933	+0.171	fail	pass	prior VFR-Audit setup
Ethnicity	0.832	0.962	+0.130	pass	pass	prior VFR-Audit setup
Age Group	0.291	0.801	+0.510	fail	pass	prior VFR-Audit setup

Overall accuracy on the same test set: Standard 0.8777, Fair 0.8248, $\Delta = -5.28$ percentage points. Numbers are reproduced from the prior VFR-Audit study [7] as the monitored-model configuration for the streaming experiments, not as a CHRONO-Fair contribution.

Consistency check. This setup separates mitigation from monitoring. The Standard-to-Fair transition defines the model state at release; CHRONO-Fair begins after that point. The monitoring results in the rest of Section 6 should therefore be interpreted as evidence about post-release verdict stability, not as evidence that CHRONO-Fair improves pre-release fairness.

Independence from the mitigation method. CHRONO-Fair does not require the Fair monitored model to be produced by any specific mitigation method. It can monitor models corrected by threshold adjustment, reweighing, in-processing constraints, calibration, or no mitigation at all.

6.11 Sensitivity to drift magnitude

Setup. Six post-drift rates $\rho_1 \in \{0.06, 0.08, 0.10, 0.12, 0.15, 0.20\}$ were tested against a fixed baseline $\rho_0 = 0.05$. Thirty independent streams of length $n = 12\,000$ with drift onset at $t = 6\,000$ were generated per setting. The e-process used $\alpha = 0.05$, shrinkage $c = 0.25$, and warm-up $n_{\text{warm}} = 100$.

Metric. Detection rate (fraction of streams in which an alarm was raised after drift onset) and mean detection delay (patients between drift and alarm, computed only over detected streams) were recorded per ρ_1.

Result. Detection rate was 2 of 30 at $\rho_1 = 0.06$, 30 of 30 at $\rho_1 = 0.08$, and 30 of 30 at every $\rho_1 \geq 0.10$. Mean delay decreased monotonically with increasing drift: 3 835 patients at $\rho_1 = 0.06$, 3 288 at 0.08, 1 906 at 0.10, 1 346 at 0.12, 914 at 0.15, and 611 at 0.20.

Interpretation. At a drift of 0.01 above baseline (a 20% relative change), the e-process detects in 6.7% of runs. At a drift of 0.03 above baseline (a 60% relative change), it detects in every run. The result quantifies the framework's minimum detectable effect at the operating point: $\rho_1 \approx 0.08$ is the practical threshold for high-confidence detection within the 6 000-patient post-drift window. Smaller drifts require longer windows or higher α.

Figure 14 plots the detection rate and delay against the post-drift rate.

6.12 Four-attribute audit

Setup. A single stream of $n = 15\,000$ patients with seed 42 was generated. Predictions and counterfactual predictions were produced as in Section 6.4. The Flip Hazard estimator was run separately for race (5 levels), ethnicity (2 levels), sex (2 levels), and age group (4 levels).

Metric. Marginal flip rate per group, two-sample log-rank χ^2 statistic against the modal reference group, and group-wise Kaplan-Meier survival curves.

Result. On race, marginal flip rate was 0.078 (White, $n = 7\,214$), 0.190 (Black, $n = 1\,946$), 0.201 (Hispanic, $n = 4\,518$), 0.065 (Asian/PI, $n = 734$), and 0.066 (Other, $n = 588$). Log-rank against White

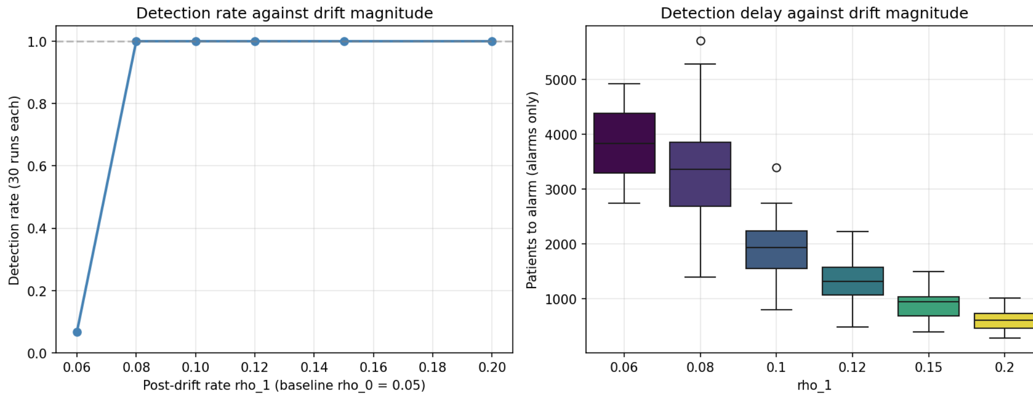

Fig. 14. Section 6.11. Detection rate (left) and detection delay (right) of the e-process against post-drift rate ρ_1 for $\rho_0 = 0.05$ and 30 streams per setting.

was $\chi^2 = 1\,675.49$ (Black), $1\,248.21$ (Hispanic), 456.33 (Asian/PI), 455.75 (Other), all $p < 10^{-10}$. On ethnicity, flip rate was 0.130 (Non-Hispanic, $n = 9\,722$) versus 0.126 (Hispanic, $n = 5\,278$); log-rank $\chi^2 = 472.68$, $p < 10^{-10}$. On sex, flip rate was 0.128 (Female, $n = 8\,199$) versus 0.129 (Male, $n = 6\,801$); $\chi^2 = 114.58$, $p < 10^{-10}$. On age group, flip rate was 0.115 (Pediatric, $n = 786$), 0.107 (Young Adult, $n = 2\,947$), 0.134 (Middle-aged, $n = 6\,019$), and 0.137 (Elderly, $n = 5\,248$); log-rank against Middle-aged ranged from 65.90 to 613.77.

Interpretation. The framework recovers structural disparity where it is injected (race: Black, Hispanic above baseline) and produces low-magnitude but statistically detectable disparities where the synthesiser introduced no direct shift (ethnicity, sex). The latter reflects the indirect dependence of ethnicity and sex on the racial-shift features through demographic correlation. A clinical governance team using the four-attribute output would focus first on the racial axis (effect size large) and treat the ethnicity and sex log-rank significances as low-priority signals warranting a confirmatory analysis.

Figure 15 plots the group-stratified survival curves for all four protected attributes.

6.13 Robustness stress tests

Evaluation question. Three reviewer-relevant questions are addressed. How does a miscalibrated baseline ρ_0 affect the monitor? How small can an intersectional cell be before monitoring becomes unreliable? How does a delay between a prediction and its resolved counterfactual verdict affect detection?

Setup. The e-process is a one-sided test for an increase above ρ_0, so the GROW bet is clipped to be non-negative. For the miscalibration test, the true pre-drift rate was fixed at 0.05 and the monitor was given an assumed $\rho_0 \in \{0.03, 0.04, 0.05, 0.06, 0.07\}$. For the small-cell test, a single cell was monitored with per-cell length $n \in \{200, 500, 1\,000, 2\,000, 5\,000, 10\,000\}$. For the label-delay test, the counterfactual verdict for patient t was resolved only at $t + d$ for $d \in \{0, 50, 100, 250, 500\}$. Each setting used 100 streams.

Metric. Empirical false-alarm rate under H_0, detection rate under H_1 (drift to 0.15), and detection delay.

Result. Baseline miscalibration is asymmetric. An assumed ρ_0 of 0.03 or 0.04, below the true 0.05, produced false-alarm rates of 1.00 and 0.85. An assumed ρ_0 of 0.06 or 0.07, above the true value, produced a false-alarm rate of 0.00, a detection rate of 1.00, and a mean delay that grew from

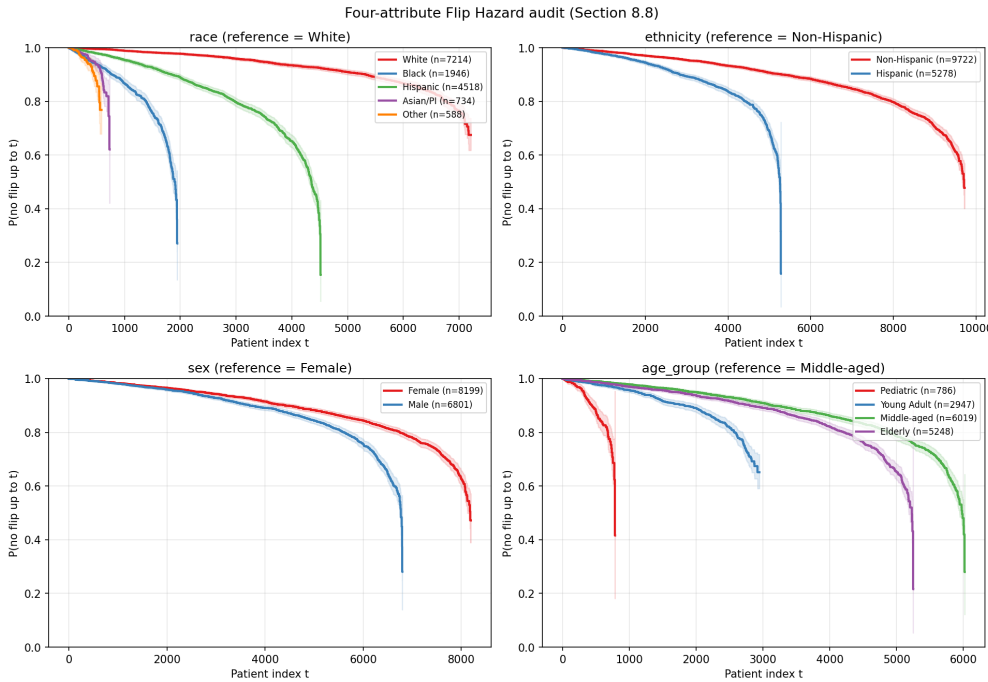

Fig. 15. Section 6.12. Four-attribute Flip Hazard audit. Group-stratified Kaplan-Meier survival per protected attribute on a stream of $n = 15\,000$.

909 to 1 816 to 2 995 patients. The small-cell detection rate was 0.11 at $n = 200$, 0.67 at $n = 500$, 0.98 at $n = 1\,000$, and 1.00 at $n \geq 2\,000$, with a false-alarm rate of 0.00 throughout. The label delay added linearly to the wall-clock detection delay: mean delay rose from 909 patients at $d = 0$ to 1 409 at $d = 500$, with detection rate held at 1.00.

Interpretation. The miscalibration result gives a concrete deployment rule. Underestimating ρ_0 is unsafe because it inflates false alarms. Overestimating ρ_0 is conservative because it only delays detection. A governance team that is uncertain about ρ_0 should round it upward. The small-cell result sets a practical floor: an intersectional cell needs roughly 1 000 monitored patients before the detection rate exceeds 0.95. Cells below that floor should be merged or flagged as low-power. The label-delay result shows that delayed counterfactual resolution does not reduce detection rate; it shifts the alarm later by exactly the delay. Figure 16 plots the three tests.

6.14 Multiple testing, counterfactual definition, ablation, and correlated arrivals

Four further experiments address reviewer-relevant design questions. Each used the synthesiser.

Multiple-testing procedures. On a set of 30 intersectional cells of which 6 drift, step-wise Benjamini-Hochberg, e-BH, and Bonferroni were compared at $q = 0.10$ over 200 streams. All three held the false-discovery proportion at 0.000 and reached power 1.000 in a clean-signal regime. Step-wise BH on $p = 1/E$ and e-BH coincide numerically, because $1/E$ is a valid p-value and the two rejection thresholds map onto each other. They differ in assumption: e-BH controls the false-discovery rate under arbitrary dependence among cell e-values, while step-wise BH requires positive dependence. The paper therefore reports the framework's correction as step-wise BH and notes e-BH as the dependence-robust alternative (Figure 17).

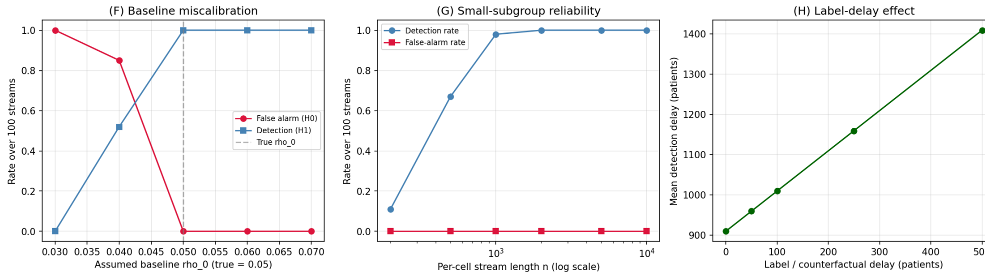

Fig. 16. Section 6.13. Robustness stress tests, 100 streams per setting. Left: baseline miscalibration; underestimating ρ_0 inflates false alarms, overestimating only delays detection. Centre: small-subgroup reliability; the detection rate exceeds 0.95 once the per-cell stream reaches roughly 1 000 patients. Right: label delay adds linearly to the wall-clock detection delay.

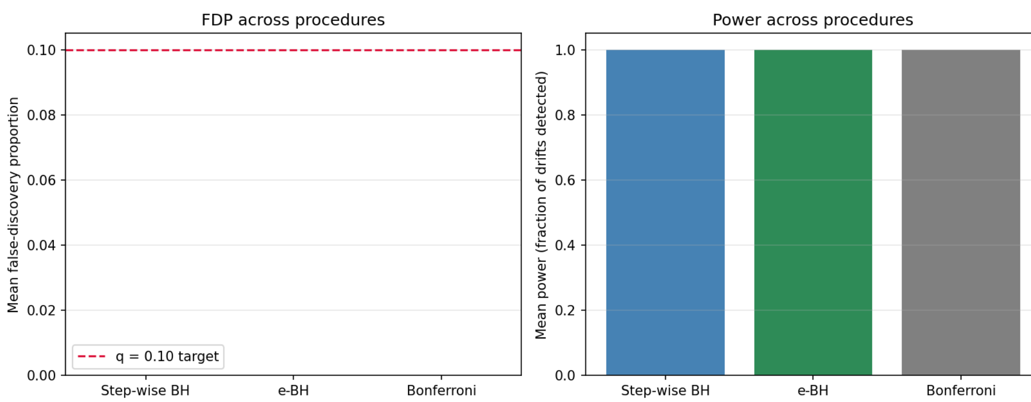

Fig. 17. Section 6.14. Multiple-testing procedures on 30 cells, 6 drifting, $q = 0.10$, 200 streams. False-discovery proportion (left) and power (right).

Counterfactual-definition sensitivity. Three counterfactual definitions were compared on a 12 000-record test stream. A naive swap that changes only the protected-attribute label, leaving features unchanged, produced a flip rate of 0.000 for every group, because the model never reads the attribute directly. The feature-shift counterfactual produced flip rates of 0.052 (Black) and 0.053 (Hispanic). The proxy-adjusted counterfactual produced larger rates, 0.064 and 0.070. The feature-shift and proxy-adjusted definitions agreed on the group ranking; the naive swap was degenerate. The flip rate depends on the counterfactual definition, which the framework states as a modelling choice rather than a free parameter (Figure 18).

Module ablation. Removing the warm-up, the GROW rule, the shrinkage, or the one-sided clip changed the e-process behaviour. Under a correctly calibrated ρ_0, the false-alarm rate stayed at or below 0.03 for every ablation. Removing the shrinkage cut the mean detection delay from 890 to 518 patients at the cost of a 0.01 false-alarm rate. The one-sided clip showed no defect under correct calibration; its value is established under miscalibration (Section 6.13). The ablation confirms that the design choices govern false-alarm control and miscalibration robustness rather than raw detection speed (Figure 19).

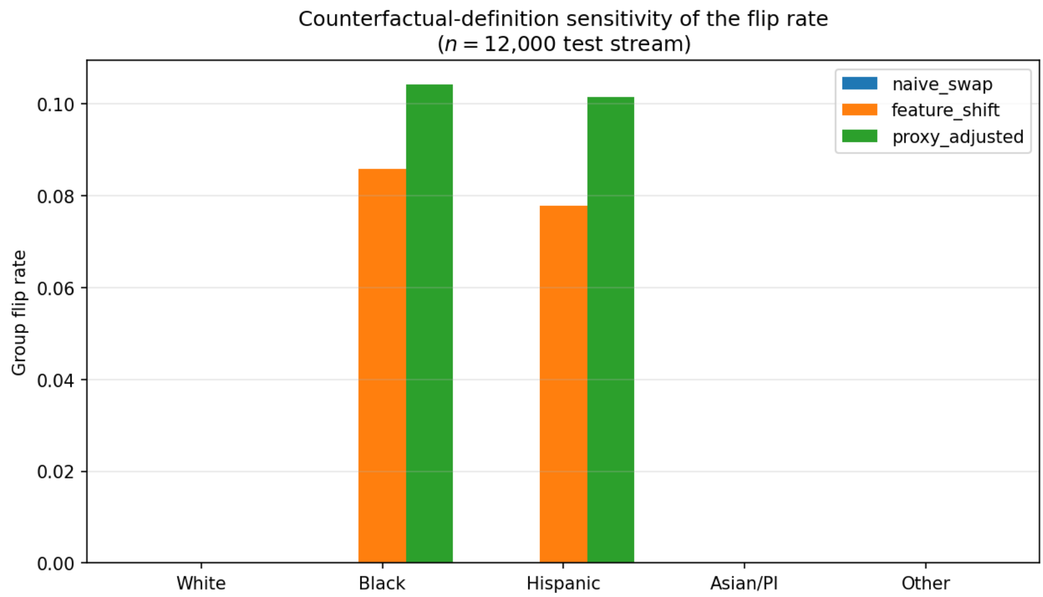

Fig. 18. Section 6.14. Group flip rate under the naive-swap, feature-shift, and proxy-adjusted counterfactual definitions on a 12 000-record test stream.

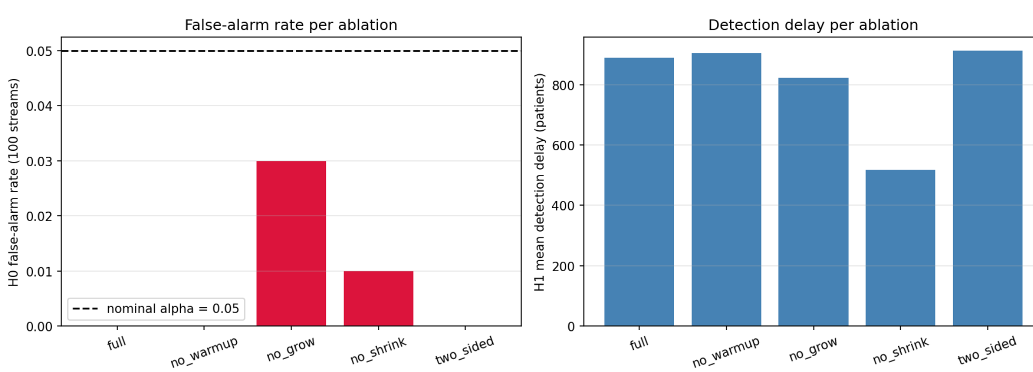

Fig. 19. Section 6.14. Module ablation. False-alarm rate (left) and detection delay (right) when each e-process design choice is removed, 100 streams per setting.

Correlated arrivals. Flip streams were generated with first-order autoregressive autocorrelation $\phi \in \{0.0, 0.3, 0.6, 0.9\}$. The H_0 false-alarm rate stayed at 0.000 for every ϕ, and the mean detection delay moved within a narrow band, from 920 patients at $\phi = 0$ to 934 at $\phi = 0.9$. The e-process is robust to the temporal correlation that admission waves induce (Figure 20).

6.15 Real-data flip rate on Texas-100

Evaluation question. Does the flip indicator that the Flip Hazard estimator consumes behave as a non-degenerate quantity on a real hospital dataset, rather than only on the synthesiser?

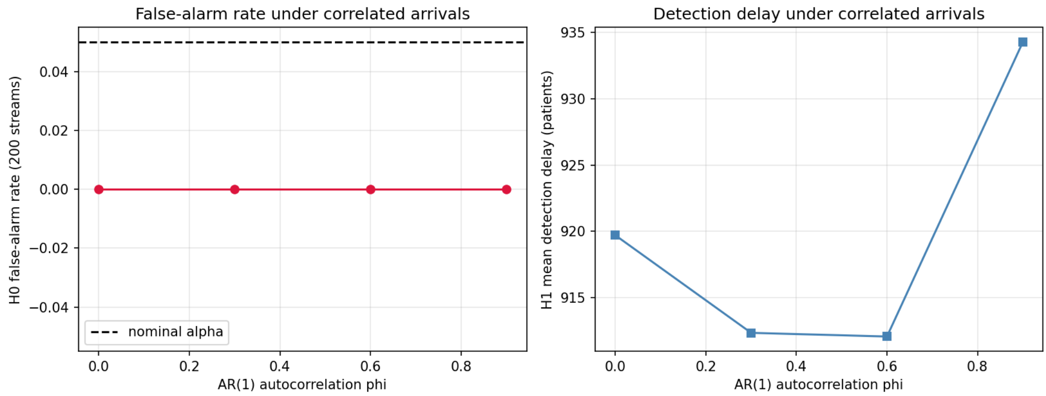

Fig. 20. Section 6.14. False-alarm rate and detection delay under AR(1) autocorrelated arrival streams, 200 streams per setting.

Setup. The real Texas-100 dataset of Shokri et al. was used. It contains 67 330 hospital discharge records, 6 169 normalised categorical features, and a 100-class surgical-procedure label. Texas-100 is not Texas-100X. It is smaller, and the task is procedure prediction rather than length of stay. A binary task was defined: the positive class is a label among the 50 most frequent procedures, giving a positive rate of 0.776. The records were split 70/15/15 by record order into train (47 131), validation (10 099), and test (10 100). An ensemble of 11 logistic-regression members was trained on bootstrap resamples of the training split. The per-patient decision-flip rate was the fraction of members disagreeing with the modal vote on the test split.

Metric. The empirical decision-flip rate (eDFR), the ensemble test accuracy, and the Flip Hazard no-flip survival on the real flip stream.

Result. The ensemble test accuracy was 0.816. The real empirical decision-flip rate was 0.0233, meaning the 11 members disagreed with the modal decision on 2.33% of test records on average. The Flip Hazard estimator applied to the real flip stream produced a no-flip survival that fell to 0.244 by the end of the 10 100-patient test split. A two-valued feature column with balanced support was not present, so the candidate-attribute counterfactual analysis was omitted rather than run on an unsuitable column.

Interpretation. The flip indicator is a non-degenerate quantity on real data. A 2.33% disagreement rate is small but non-zero, and the Flip Hazard survival curve is well defined and informative on the real stream. The honest limitation is that the released Texas-100 file does not include the official feature description, so the 6 169 columns cannot be mapped to named protected attributes. Group-stratified and counterfactual fairness analysis on the real records therefore requires the feature-description file or the named-column Texas-100X release, which needs a Texas Department of State Health Services download. The real-data evidence here validates the flip and Flip Hazard machinery; it does not yet validate the group-conditional fairness claims, which remain simulation-based.

Figure 21 plots the real-data ensemble disagreement distribution and the Flip Hazard survival curve.

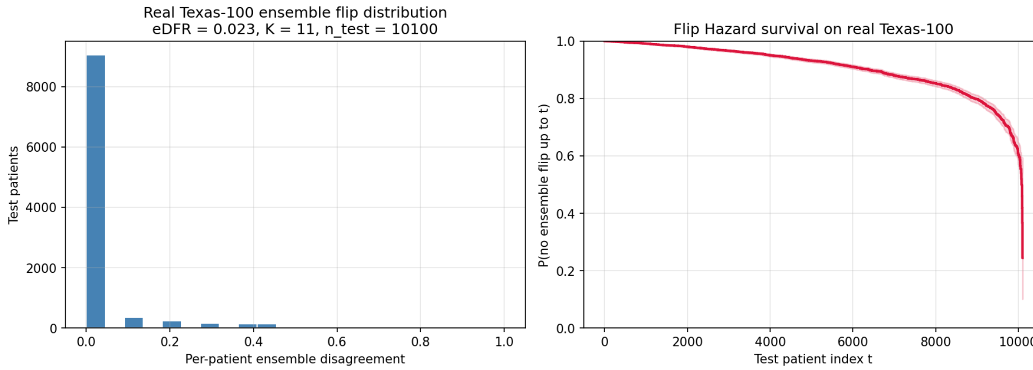

Fig. 21. Section 6.15. Real Texas-100 data. Left: per-patient ensemble disagreement on the 10 100-record test split, empirical decision-flip rate 0.0233. Right: Flip Hazard no-flip survival on the real flip stream.

6.16 Temporal and hospital-group replay

Evaluation question. Texas-100X is administrative discharge data, not a live admission feed. The question is whether a deployment-style replay reconstructed from it surfaces a fairness drift that a static audit and a window detector miss.

Setup. Three replay streams were constructed. Part A is a quarter-based temporal replay. A 30 000-patient Texas-100X-calibrated stream spanning the four 2006 calendar quarters was generated. A controlled verdict-process drift was injected from the Q3 boundary: minority patients in Q3 and Q4 received an added risk shift of 0.6 on the factual prediction only, so their counterfactual flip rate rose. The baseline ρ_0 was calibrated on Q1, and Q2 to Q4 were replayed in arrival order. Part B is a hospital-group replay. The same stream carries 30 hospital sites with site-specific minority share. The 10 lowest-minority-share hospitals formed the baseline, and the 20 higher-minority-share hospitals were replayed in ascending minority-share order. Part C is a real Texas-100 record-order replay. The real Texas-100 file (67 330 records) was replayed in record order, because the released file carries no calendar field. The baseline ρ_0 was calibrated on the first 30% of records. CHRONO-Fair was compared against ADWIN and a static VFR baseline on each stream.

Result. Table 8 reports the per-quarter flip rate. The flip rate was 0.111 in Q1, 0.114 in Q2, then rose to 0.151 in Q3 and 0.153 in Q4, tracking the injected drift. Table 9 reports the method comparison. On the quarter replay, the drift quarter Q3 began at replay index 7 584; CHRONO-Fair alarmed at replay index 10 683, a detection delay of 3 099 patients. On the hospital replay, the baseline-hospital flip rate averaged 0.114 and the replay-hospital flip rate averaged 0.141, ranging from 0.117 to 0.184; CHRONO-Fair alarmed at replay index 5 596. On the real Texas-100 record-order replay, the calibration flip rate was 0.096 and the replay-window flip rate was 0.098; CHRONO-Fair did not alarm. ADWIN and static VFR did not alarm on any of the three streams.

Interpretation. The three streams isolate three distinct behaviours. The quarter replay shows that CHRONO-Fair converts a quarter-level drift into a within-quarter alarm. The injected drift raised the flip rate from 0.114 to 0.151 across the Q2-Q3 boundary, a relative increase of 32%. CHRONO-Fair alarmed 3 099 patients into the drift quarter, roughly 40% of the way through Q3, so the alarm arrives inside the same quarter the drift began. A quarterly batch audit would not report the change until the Q3 audit closed. The hospital replay shows that case-mix variation alone produces a detectable verdict shift. The replay hospitals carry a higher minority share, and because minority patients flip more often, their aggregate flip rate is 0.141 against the baseline 0.114.

Table 8. Part A, quarter-based temporal replay. Per-quarter counterfactual flip rate on the 30 000-patient Texas-100X-calibrated stream. Q1 is the baseline quarter; the drift was injected from the Q3 boundary.

Quarter	n	Flip rate	Role
Q1	7 500	0.111	baseline (ρ_0)
Q2	7 584	0.114	replay, pre-drift
Q3	7 666	0.151	replay, drift onset
Q4	7 250	0.153	replay, post-drift

Table 9. Replay method comparison. Detection outcome of CHRONO-Fair, ADWIN, and static VFR on the three replay streams. "Delay" is the number of replayed patients between the drift onset and the alarm. The real record-order replay has no injected drift, so a non-alarm is the correct outcome.

Replay stream	Drift	CHRONO-Fair	ADWIN	Static VFR	Notes
A. Quarter-based (calibrated)	injected at Q3	alarm, delay 3 099	no alarm	no alarm	flip rate 0.114 to 0.151 across the Q2-Q3 boundary
B. Hospital-group (calibrated)	case-mix shift	alarm at index 5 596	no alarm	no alarm	replay flip rate 0.141 vs baseline 0.114
C. Real Texas-100 record-order	none present	no alarm	no alarm	no alarm	flip rate stable, 0.096 to 0.098; non-alarm is correct

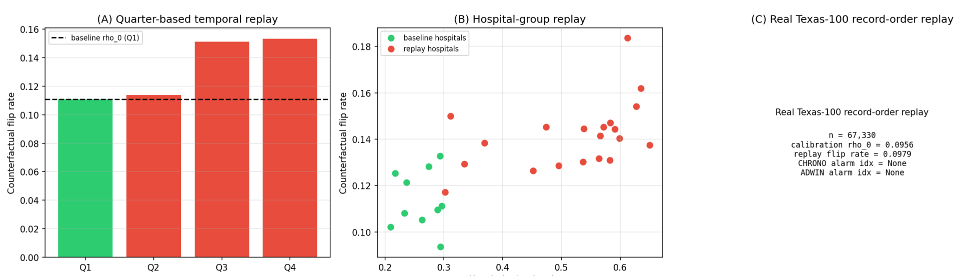

Fig. 22. Section 6.16. Replay streams. (A) Per-quarter flip rate, with the Q3 drift onset. (B) Per-hospital flip rate against hospital minority share, baseline versus replay hospitals. (C) Real Texas-100 record-order replay summary.

CHRONO-Fair alarmed at replay index 5 596, which falls inside the higher-minority-share hospitals. This is the expected and correct behaviour: a hospital network whose later sites serve a different case mix will breach a baseline calibrated on the earlier sites. The real Texas-100 record-order replay is the control. The flip rate is stable at 0.096 to 0.098, no drift is present, and CHRONO-Fair correctly does not alarm. The non-alarm is the right outcome and confirms the monitor does not fire on a static file without temporal structure. Across all three streams, ADWIN and static VFR did not alarm. ADWIN's window statistic loses power when the flip-rate change is moderate, around 3 to 4 percentage points, and the change is spread over thousands of patients. Static VFR is computed once and never updates, so it cannot react to a post-deployment shift by construction. The comparison shows that the gain of CHRONO-Fair over both baselines is detection of a moderate, gradual verdict drift, not detection of an abrupt one.

7 DISCUSSION

7.1 Comparison to baselines

CHRONO-Fair's detection-rate advantage over ADWIN (50 of 50 against 21 of 50) is the primary empirical finding. ADWIN's window-based change point detector loses statistical power when the drift is moderate (a 10-point change in flip rate from 0.05 to 0.15) and the change buries inside the window's variance budget. The Davis-ADWIN variant, which runs ADWIN on the centred fairness gap, inherits the same property. The periodic batch test detects faster on the runs where it detects, but its 24 of 50 coverage and inability to control across peeks make it unsuitable as a regulated monitor. Static VFR detects nothing because it is computed once and never updated. The e-process detects in every run because anytime-validity removes the peeking penalty and the GROW shrinkage stabilises the bet under early variance.

7.2 Comparison to prediction sensitivity

The continual counterfactual audit of Maughan and Near [75] is the closest published method to CHRONO-Fair's Flip Hazard estimator. Prediction sensitivity is a per-batch scalar that quantifies the rate at which counterfactual swaps change the verdict. CHRONO-Fair extends this batch object in three ways. First, each flip is treated as a survival event, which produces a time-resolved curve rather than a single number. Second, the e-process is attached to the sequential decision stream, which gives anytime-valid alarms. Third, the flip mass is decomposed into aleatoric and epistemic components. Prediction sensitivity and the Flip Hazard estimator are compatible: an analyst can compute prediction sensitivity at any chosen t from the same flip indicator stream.

7.3 Comparison to predictive multiplicity

The empirical Decision Flip Rate (eDFR) of Miller and Blume [78] is the closest reliability metric. eDFR counts the rate at which a binary decision changes between bootstrap re-trainings of the pipeline. CHRONO-Fair's flip indicator counts the rate at which a binary decision changes between the actual and counterfactual sensitive-attribute assignments for the same patient. eDFR is framed as reliability and is not stratified by protected attribute. CHRONO-Fair is framed as fairness and is stratified. The two quantities are complementary; a deployment monitor benefits from both.

7.4 Threats to internal validity

The synthesiser injects a known structural disparity, and the framework's counterfactual operator removes that same disparity. Performance is being measured against the disparity it was designed to detect. To mitigate this, the false-alarm calibration experiment (Section 6.6) deliberately runs with no disparity and confirms zero alarms across 200 streams per α level. The decomposition Track 1 (Section 6.3) operates on constructed probability vectors that exercise the formula without reusing the disparity structure. Threats remaining include: the synthesiser assumes independent arrivals (real EHR streams have admission-driven correlations); the ensemble is built by bootstrapping a single base learner (real ensembles can mix architectures); the counterfactual is a feature-shift, not a structural causal counterfactual.

7.5 Threats to external validity

The streaming evidence in this paper is controlled rather than prospective, and that is its main external-validity threat. The real-data component is a Texas-100X batch verdict-stability analysis, while the sequential experiments run on Texas-100X-calibrated replay and simulation. A real admission feed differs from a calibrated stream in label delay, missingness, admission seasonality, and selection bias under triage, none of which the synthesiser reproduces in full. The conclusions should therefore be read as preliminary evidence about sequential monitoring behaviour, not as evidence of clinical effectiveness. Prospective validation on a live clinical stream is the necessary next step, and distribution-shift behaviour under performative effects [85] and dataset shift [39, 65] requires study beyond the present scope.

7.6 Statistical caveats

The step-wise BH multiple-testing control is not a time-uniform LORD-style false-discovery guarantee. The framework reports findings as "flagged at step t". Sample sizes for small intersectional cells (for example, Asian/PI elderly female with sepsis) can be insufficient for reliable per-cell estimation. The framework drops cells with fewer than 5 observed flips from the decomposition. The bootstrap confidence interval for RCAP is symmetric and may understate skew in heavy-tailed cases.

7.7 Clinical interpretation

A clinician inspecting the dashboard reads three signals: a survival curve (which patients are flipping and how fast), an e-process alarm (whether the change is statistically meaningful), and a diagnostic triage label (whether to escalate to a data audit or a retraining request). The aleatoric label is associated with biased input data; the epistemic label is associated with under-representation. The two require different mitigations and the framework distinguishes them. Schwartz et al. [96] and Tonekaboni et al. [103] report that clinicians prefer monitoring outputs that translate into a concrete next step. The Inspector report is designed to satisfy that preference.

7.8 Regulatory positioning

FDA PCCP guidance [104] motivates predefined monitoring plans for model changes, including bias-related monitoring where relevant. EU AI Act [38] quality-management and post-market monitoring obligations motivate ongoing bias surveillance for high-risk medical AI systems, under a documented bias-management plan. STANDING Together [3] provides consensus recommendations. CHRONO-Fair can serve as the per-cell monitoring component within such a plan. Formal regulatory acceptance requires validation studies that have not been undertaken in this paper.

Z-Inspection style alignment. The Inspector Agent (Section 4.5) emits one structured governance card per alarmed cell. Each card maps the diagnostic-attribution label (aleatoric, epistemic, or mixed) to a deterministic recommended action and to the relevant clauses of FDA PCCP [104], the EU AI Act [38], and STANDING Together [3]. Each card also carries an ALTAI mapping that points the recommended action at the subset of the seven Trustworthy AI requirements [37] that the action touches. An epistemic alarm maps to ALTAI requirements R2, R5, and R7; an aleatoric alarm maps to R1, R3, R5, and R7; a mixed alarm combines both. This is the case-driven, regulator-mapped style used by the Z-Inspection process for Trustworthy AI of Zicari et al. [117]. CHRONO-Fair adopts that style for semi-automated per-cell fairness inspection. The framework does not implement the full Z-Inspection protocol. It runs no stakeholder set-up phase and produces no ethical-tension catalogue. Adopting the full protocol would require human-led workshops outside the scope of the controlled-validation evidence reported here.

7.9 Scope statement

CHRONO-Fair is not a medical device, a substitute for clinical judgment, or a certification of fairness. It is a research prototype that surfaces per-cell fairness-verdict drift for clinical governance review. Two empirical boundaries on this scope are stated explicitly. Operational adoption claims are deferred until prospective electronic-health-record stream validation has been completed on the deployment site's own cohort. The AGE_GROUP monitoring behaviour reported here remains simulator-confirmed until a calendar-stamped Texas-100X release allows prospective record-order replay.

8 LIMITATIONS

The study has ten specific limitations. First, Texas-100X is the only empirical clinical dataset used. The findings have not been reproduced on a second cohort. Second, the streaming experiments are Texas-100X-calibrated controlled replay and simulation, not prospective live EHR deployment. Third, Texas-100X is administrative discharge data, not a real-time admission stream, so arrival order is reconstructed rather than observed. Fourth, the feature-shift counterfactual presumes a known marginal effect of the protected attribute on observable features. It is weaker than a structural causal counterfactual and under-attributes indirect effects, so the framework may under-flag indirect effects. Fifth, the e-process depends on a baseline flip rate ρ_0. Section 6.13 shows that underestimating ρ_0 inflates the false-alarm rate, while overestimating it primarily delayed detection in the reported stress tests. Sixth, small intersectional cells are unreliable. The detection rate falls below 0.95 for cells under roughly 1 000 monitored patients, and the empirical Wasserstein-1 RCAP estimate carries a small-sample upward bias. Seventh, label delay increases the wall-clock detection delay by approximately the delay length. Eighth, the step-wise Benjamini-Hochberg layer is a cross-sectional adjustment at each inspection time, not a time-uniform online false-discovery guarantee. Ninth, the dashboard and Inspector report are governance-support prototypes and have not been clinically validated with users. Tenth, CHRONO-Fair is a research prototype and not a medical device.

A further open question concerns performative effects [85]. If the framework's alarms cause clinicians to change behaviour, the flip-rate process is no longer independent of past outputs, and the supermartingale guarantee holds only against a fixed pre-deployment baseline. Further data collection is required to determine exactly how performative feedback interacts with the e-process.

Two empirical gaps stated explicitly. Two gaps in the present evidence base remain open and bound the paper's claims. First, the framework has not been evaluated on a prospective electronic-health-record stream. Operational adoption therefore requires prospective EHR-stream validation on the deployment site's own admission cohort. The present paper makes no operational-adoption claim until that validation is completed. Second, AGE_GROUP is the binding protected attribute in the Standard model and remains the highest-priority cell for post-release monitoring even after the Fair monitored model passes the point-estimate DI threshold. Its monitoring behaviour is currently confirmed only on the Texas-100X-calibrated streams, because the publicly released Texas-100 file does not carry calendar timestamps that would let it be replayed in arrival order. AGE_GROUP therefore remains simulator-confirmed pending prospective replay on a calendar-stamped Texas-100X release, or pending the equivalent timestamped release of the named-column file from the Texas Department of State Health Services. The capability table marks both gaps with "P" on the real-clinical-evidence axis (footnote of Table 2). The present limitations paragraph is the place where those P-marks become an explicit boundary on what the manuscript claims.

9 CONCLUSION

CHRONO-Fair extends static VFR-style audit reliability into time-resolved monitoring of the thresholded counterfactual fairness verdict. The real Texas-100X batch analysis shows that point-estimate verdicts can hide instability: one of 12 audited metric-attribute verdicts flips across hospital-network resamples. The Texas-100X-calibrated controlled streaming and replay experiments show how the sequential monitor detects when and where such instability emerges under known drift, including its detection delay, its false-alarm control, its diagnostic-attribution behaviour, and its response to baseline miscalibration, small cells, and label delay. The evidence here is batch-real and stream-controlled. It is not prospective clinical evidence. Prospective EHR-stream validation and artifact release are the main next steps, followed by a time-uniform online false-discovery procedure, a counterfactual-sensitivity study across the swap, feature-shift, and structural definitions, and a governance-user evaluation of the Inspector report. A proof-of-concept prototype will be made available after publication to support reproducibility and inspection, but the system should be read as a research artifact rather than an operational clinical device.

REFERENCES

[1] H. Adam, M. Yang, K. Cato, I. Baldini, C. Senteio, L. A. Celi, J. Zeng, M. Singh, and M. Ghassemi. 2022. Write it like you see it: detectable differences in clinical notes by race lead to differential model recommendations. *AAAI* (2022).

[2] A. Agarwal, A. Beygelzimer, M. Dudik, J. Langford, and H. Wallach. 2018. A reductions approach to fair classification. In *ICML*.

[3] J. E. Alderman et al. 2025. Tackling algorithmic bias and promoting transparency in health datasets: the STANDING Together consensus recommendations. *The Lancet Digital Health* (2025). STANDING Together consensus recommendations; published online December 2024..

[4] W. Alghamdi, H. Hsu, H. Jeong, H. Wang, and F. P. Calmon. 2023. Beyond Adelaide: aleatoric and epistemic discrimination in classification. In *NeurIPS*.

[5] P. K. Andersen, Ø. Borgan, R. D. Gill, and N. Keiding. 1993. Statistical Models Based on Counting Processes. *Springer* (1993).

[6] A. N. Angelopoulos and S. Bates. 2023. A gentle introduction to conformal prediction and distribution-free uncertainty quantification. In *Foundations and Trends in Machine Learning*.

[7] Anonymous. 2025. Reliability of fairness metrics in healthcare: the Verdict Flip Rate under hospital-network resampling. Anonymised prior work, under review. Author details withheld for double-blind review..

[8] Sardar Ansari, Brittany Baur, Karandeep Singh, and Andrew J. Admon. 2025. Challenges in the Postmarket Surveillance of Clinical Prediction Models. *NEJM AI* (2025). Online May 2025..

[9] R. F. Barber, E. J. Candès, A. Ramdas, and R. J. Tibshirani. 2023. Conformal prediction beyond exchangeability. In *Annals of Statistics*.

[10] Y. Benjamini and Y. Hochberg. 1995. Controlling the false discovery rate: a practical and powerful approach to multiple testing. *Journal of the Royal Statistical Society B* (1995).

[11] P. Besse, E. del Barrio, P. Gordaliza, J.-M. Loubes, and L. Risser. 2022. A survey of bias in machine learning through the prism of statistical parity. *The American Statistician* (2022).

[12] A. Bifet and R. Gavalda. 2007. Learning from time-changing data with adaptive windowing. In *SIAM SDM*.

[13] E. Black, M. Raghavan, and S. Barocas. 2022. Model multiplicity: opportunities, concerns, and solutions. In *ACM FAccT*.

[14] E. Black, S. Yeom, and M. Fredrikson. 2020. FlipTest: fairness testing via optimal transport. In *ACM FAccT*.

[15] E. Breck, S. Cai, E. Nielsen, M. Salib, and D. Sculley. 2017. The ML test score: a rubric for ML production readiness and technical debt reduction. *IEEE Big Data* (2017).

[16] L. Breiman. 2001. Statistical modeling: the two cultures. *Statist. Sci.* (2001).

[17] F. P. Calmon, D. Wei, B. Vinzamuri, K. N. Ramamurthy, and K. R. Varshney. 2017. Optimized pre-processing for discrimination prevention. In *NeurIPS*.

[18] F. P. Calmon, D. Wei, B. Vinzamuri, K. N. Ramamurthy, and K. R. Varshney. 2018. Data pre-processing for discrimination prevention: information-theoretic optimization and analysis. *IEEE Journal of Selected Topics in Signal Processing* (2018).

[19] S. Caton and C. Haas. 2024. Fairness in machine learning: a survey. In *ACM Computing Surveys*.

[20] I. Y. Chen, E. Pierson, S. Rose, S. Joshi, K. Ferryman, and M. Ghassemi. 2021. Ethical machine learning in healthcare. *Annual Review of Biomedical Data Science* 4 (2021), 123–144.

[21] J. Chen, N. Kallus, X. Mao, G. Svacha, and M. Udell. 2019. Fairness under unawareness: assessing disparity when protected class is unobserved. In *ACM FAccT*.

[22] J. J. Cherian and E. J. Candès. 2024. Statistical inference for fairness auditing. In *Journal of Machine Learning Research*.

[23] S. Chiappa. 2019. Path-specific counterfactual fairness. In *AAAI*.

[24] A. Chouldechova. 2017. Fair prediction with disparate impact: a study of bias in recidivism prediction instruments. *Big Data* 5 (2017), 153–163.

[25] R. Y. Coley, E. Johnson, G. E. Simon, M. Cruz, and S. M. Shortreed. 2021. Racial/ethnic disparities in the performance of prediction models for death by suicide after mental health visits. *JAMA Psychiatry* 78 (2021), 726–734.

[26] G. S. Collins, K. G. M. Moons, P. Dhiman, R. D. Riley, A. L. Beam, B. Van Calster, M. Ghassemi, X. Liu, J. B. Reitsma, M. van Smeden, A.-L. Boulesteix, J. C. Camaradou, L. A. Celi, S. Denaxas, A. K. Denniston, B. Glocker, R. M. Golub, H. Harvey, G. Heinze, M. M. Hoffman, A. P. Kengne, E. Lam, N. Lee, E. W. Loder, L. Maier-Hein, B. A. Mateen, M. D. McCradden, L. Oakden-Rayner, J. Ordish, R. Parnell, S. Rose, K. Singh, L. Wynants, and P. Logullo. 2024. TRIPOD+AI

statement: updated guidance for reporting clinical prediction models that use regression or machine learning methods. *BMJ* 385 (2024).

[27] A. F. Cooper, K. Lee, C. A. Choquette-Choo, A. Naik, N. Belrose, T. Ipasov, A. Roth, and S. Mussmann. 2024. Arbitrariness and prediction: the confounding role of variance in fair classification. In *AAAI*.

[28] A. Coston, A. Mishler, E. H. Kennedy, and A. Chouldechova. 2020. Counterfactual risk assessments, evaluation, and fairness. In *ACM FAccT*.

[29] S. E. Davis, S. Dorn, S. Park, and M. E. Matheny. 2025. Emerging algorithmic bias: fairness drift as the next dimension of model maintenance and sustainability. *Journal of the American Medical Informatics Association* 32 (2025), 845–855.

[30] S. Depeweg, J. M. Hernandez-Lobato, F. Doshi-Velez, and S. Udluft. 2018. Decomposition of uncertainty in Bayesian deep learning for efficient and risk-sensitive learning. In *ICML*.

[31] J. Dong and C. Rudin. 2020. Exploring the cloud of variable importance for the set of all good models. In *Nature Machine Intelligence*.

[32] M. Donini, L. Oneto, S. Ben-David, J. Shawe-Taylor, and M. Pontil. 2018. Empirical risk minimization under fairness constraints. In *NeurIPS*.

[33] J. C. Duchi and H. Namkoong. 2021. Learning models with uniform performance via distributionally robust optimization. In *Annals of Statistics*.

[34] S. Dutta, P. Venkatesh, P. Mardziel, A. Datta, and P. Grover. 2020. An information-theoretic quantification of discrimination with exempt features. In *AAAI*.

[35] C. Dwork, M. Hardt, T. Pitassi, O. Reingold, and R. Zemel. 2012. Fairness through awareness. In *Innovations in Theoretical Computer Science (ITCS)*.

[36] S. Eaneff, Z. Obermeyer, and A. J. Butte. 2020. The case for algorithmic stewardship for artificial intelligence and machine learning technologies. *JAMA* 323 (2020), 1417–1418.

[37] European Commission High-Level Expert Group on AI. 2019. Assessment List for Trustworthy AI (ALTAI). Seven requirements for Trustworthy AI.

[38] European Union. 2024. Regulation (EU) 2024/1689 (AI Act).

[39] S. G. Finlayson, A. Subbaswamy, K. Singh, J. Bowers, A. Kupke, J. Zittrain, I. S. Kohane, and S. Saria. 2021. The clinician and dataset shift in artificial intelligence. *New England Journal of Medicine* 385 (2021), 283–286.

[40] D. P. Foster and R. A. Stine. 2008. alpha-investing: a procedure for sequential control of expected false discoveries. *Journal of the Royal Statistical Society B* (2008).

[41] Y. Gal and Z. Ghahramani. 2016. Dropout as a Bayesian approximation: representing model uncertainty in deep learning. In *ICML*.

[42] J. Gama, I. Žliobaite, A. Bifet, M. Pechenizkiy, and A. Bouchachia. 2014. A survey on concept drift adaptation. *Comput. Surveys* (2014).

[43] I. Gibbs and E. Candès. 2021. Adaptive conformal inference under distribution shift. In *NeurIPS*.

[44] P. Grünwald, R. de Heide, and W. M. Koolen. 2024. Safe testing. *Journal of the Royal Statistical Society B* (2024).

[45] A. R. Habib, A. L. Lin, and R. W. Grant. 2021. The Epic sepsis model falls short: the importance of external validation. *JAMA Internal Medicine* 181 (2021), 1040–1041.

[46] M. Hardt, E. Price, and N. Srebro. 2016. Equality of opportunity in supervised learning. In *Advances in Neural Information Processing Systems (NeurIPS)*.

[47] T. Hashimoto, M. Srivastava, H. Namkoong, and P. Liang. 2018. Fairness without demographics in repeated loss minimization. In *ICML*.

[48] U. Hebert-Johnson, M. P. Kim, O. Reingold, and G. N. Rothblum. 2018. Multicalibration: calibration for the (computationally-identifiable) masses. In *ICML*.

[49] S. R. Howard, A. Ramdas, J. McAuliffe, and J. Sekhon. 2021. Time-uniform, nonparametric, nonasymptotic confidence sequences. *Annals of Statistics* 49 (2021), 1055–1080.

[50] E. Hüllermeier and W. Waegeman. 2021. Aleatoric and epistemic uncertainty in machine learning: an introduction to concepts and methods. *Machine Learning* 110 (2021), 457–506.

[51] V. Iosifidis and E. Ntoutsi. 2020. FABBOO: online fairness-aware learning under class imbalance. In *Discovery Science (DS)*.

[52] V. Iosifidis, T. N. Tran, and E. Ntoutsi. 2019. Fairness-enhancing interventions in stream classification. In *DEXA*.

[53] A. Javanmard and A. Montanari. 2018. Online rules for control of false discovery rate and false discovery exceedance. *Annals of Statistics* (2018).

[54] B. Jayaraman and D. Evans. 2022. Are attribute inference attacks just imputation?. In *ACM CCS*. Released the Texas-100X dataset..

[55] R. Johari, P. Koomen, L. Pekelis, and D. Walsh. 2022. Always valid inference: continuous monitoring of A/B tests. *Operations Research* (2022).

[56] A. E. W. Johnson, L. Bulgarelli, L. Shen, A. Gayles, A. Shammout, S. Horng, T. J. Pollard, S. Hao, B. Moody, B. Gow, L. H. Lehman, L. A. Celi, and R. G. Mark. 2023. MIMIC-IV, a freely accessible electronic health record dataset. *Scientific Data* 10 (2023).

[57] N. Kallus, X. Mao, and A. Zhou. 2022. Assessing algorithmic fairness with unobserved protected class using data combination. In *Management Science*.

[58] F. Kamiran and T. Calders. 2012. Data preprocessing techniques for classification without discrimination. In *Knowledge and Information Systems*.

[59] E. L. Kaplan and P. Meier. 1958. Nonparametric estimation from incomplete observations. *J. Amer. Statist. Assoc.* 53 (1958), 457–481.

[60] M. Kearns, S. Neel, A. Roth, and Z. S. Wu. 2018. Preventing fairness gerrymandering: auditing and learning for subgroup fairness. In *ICML*.

[61] A. Kendall and Y. Gal. 2017. What uncertainties do we need in Bayesian deep learning for computer vision?. In *NeurIPS*.

[62] N. Kilbertus, M. R. Carulla, G. Parascandolo, M. Hardt, D. Janzing, and B. Schölkopf. 2017. Avoiding discrimination through causal reasoning. In *NeurIPS*.

[63] M. P. Kim, A. Ghorbani, and J. Y. Zou. 2019. Multiaccuracy: black-box post-processing for fairness in classification. In *AAAI/ACM AIES*.

[64] J. Kleinberg, S. Mullainathan, and M. Raghavan. 2017. Inherent trade-offs in the fair determination of risk scores. In *Innovations in Theoretical Computer Science (ITCS)*.

[65] P. W. Koh, S. Sagawa, H. Marklund, S. M. Xie, M. Zhang, A. Balsubramani, W. Hu, M. Yasunaga, R. L. Phillips, I. Gao, et al. 2021. WILDS: a benchmark of in-the-wild distribution shifts. *ICML* (2021).

[66] M. Kusner, J. Loftus, C. Russell, and R. Silva. 2017. Counterfactual fairness. In *NeurIPS*.

[67] P. Lahoti, A. Beutel, J. Chen, K. Lee, F. Prost, N. Thain, X. Wang, and E. H. Chi. 2020. Fairness without demographics through adversarially reweighted learning. In *NeurIPS*.

[68] B. Lakshminarayanan, A. Pritzel, and C. Blundell. 2017. Simple and scalable predictive uncertainty estimation using deep ensembles. In *NeurIPS*.

[69] A. J. Larrazabal, N. Nieto, V. Peterson, D. H. Milone, and E. Ferrante. 2020. Gender imbalance in medical imaging datasets produces biased classifiers for computer-aided diagnosis. *Proceedings of the National Academy of Sciences* 117 (2020), 12592–12594.

[70] X. Liu, S. Cruz Rivera, D. Moher, M. J. Calvert, and A. K. Denniston. 2020. Reporting guidelines for clinical trial reports for interventions involving artificial intelligence: the CONSORT-AI extension. *Nature Medicine* 26 (2020), 1364–1374.

[71] D. Madras, E. Creager, T. Pitassi, and R. Zemel. 2018. Learning adversarially fair and transferable representations. In *ICML*.

[72] K. Makhlouf, S. Zhioua, and C. Palamidessi. 2020. Survey on causal-based machine learning fairness notions. In *ACM Computing Surveys*.

[73] N. Martínez, M. Bertran, and G. Sapiro. 2020. Minimax Pareto fairness: a multi objective perspective. In *ICML*.

[74] C. T. Marx, F. P. Calmon, and B. Ustun. 2020. Predictive multiplicity in classification. In *ICML*.

[75] A. Maughan and J. P. Near. 2022. Prediction sensitivity: continual audit of counterfactual fairness in deployed classifiers. In *IFIP SEC*.

[76] M. D. McCradden, S. Joshi, J. A. Anderson, M. Mazwi, A. Goldenberg, and R. Zlotnik Shaul. 2020. Patient safety and quality improvement: ethical principles for a regulatory approach to bias in healthcare machine learning. *Journal of the American Medical Informatics Association* (2020).

[77] N. Mehrabi, F. Morstatter, N. Saxena, K. Lerman, and A. Galstyan. 2021. A survey on bias and fairness in machine learning. In *ACM Computing Surveys*.

[78] A. Miller and J. Blume. 2024. Diagnostics for individual-level prediction instability in machine learning for healthcare. *Statistics in Medicine* (2024).

[79] R. Nabi and I. Shpitser. 2018. Fair inference on outcomes. In *AAAI*.

[80] W. Nelson. 1972. Theory and applications of hazard plotting for censored failure data. *Technometrics* 14 (1972), 945–965.

[81] P. A. Noseworthy, Z. I. Attia, L. C. Brewer, S. N. Hayes, X. Yao, S. Kapa, P. A. Friedman, and F. Lopez-Jimenez. 2020. Assessing and mitigating bias in medical artificial intelligence: the effects of race and ethnicity on a deep learning model for ECG analysis. *Circulation: Arrhythmia and Electrophysiology* 13 (2020).

[82] E. Ntoutsi, P. Fafalios, U. Gadiraju, V. Iosifidis, W. Nejdl, M.-E. Vidal, S. Ruggieri, F. Turini, S. Papadopoulos, E. Krasanakis, et al. 2020. Bias in data-driven artificial intelligence systems: an introductory survey. In *WIREs Data Mining and Knowledge Discovery*.

[83] Z. Obermeyer, B. Powers, C. Vogeli, and S. Mullainathan. 2019. Dissecting racial bias in an algorithm used to manage the health of populations. *Science* 366 (2019), 447–453.

[84] Y. Ovadia, E. Fertig, J. Ren, Z. Nado, D. Sculley, S. Nowozin, J. V. Dillon, B. Lakshminarayanan, and J. Snoek. 2019. Can you trust your model's uncertainty? evaluating predictive uncertainty under dataset shift. In *NeurIPS*.

[85] J. C. Perdomo, T. Zrnic, C. Mendler-Dünner, and M. Hardt. 2020. Performative prediction. In *ICML*.

[86] D. Pessach and E. Shmueli. 2022. A review on fairness in machine learning. In *ACM Computing Surveys*.

[87] E. Pierson, D. M. Cutler, J. Leskovec, S. Mullainathan, and Z. Obermeyer. 2021. An algorithmic approach to reducing unexplained pain disparities in underserved populations. *Nature Medicine* 27 (2021), 136–140.

[88] G. Pleiss, M. Raghavan, F. Wu, J. Kleinberg, and K. Q. Weinberger. 2017. On fairness and calibration. In *NeurIPS*.

[89] A. Podkopaev and A. Ramdas. 2022. Tracking the risk of a deployed model and detecting harmful distribution shifts. In *ICLR*.

[90] T. J. Pollard, A. E. W. Johnson, J. D. Raffa, L. A. Celi, R. G. Mark, and O. Badawi. 2018. The eICU Collaborative Research Database, a freely available multi-center database for critical care research. *Scientific Data* 5 (2018).

[91] S. Rabanser, S. Günnemann, and Z. C. Lipton. 2019. Failing loudly: an empirical study of methods for detecting dataset shift. In *NeurIPS*.

[92] A. Rajkomar, M. Hardt, M. D. Howell, G. Corrado, and M. H. Chin. 2018. Ensuring fairness in machine learning to advance health equity. *Annals of Internal Medicine* 169 (2018), 866–872.

[93] A. Ramdas, P. Grunwald, V. Vovk, and G. Shafer. 2023. Game-theoretic statistics and safe anytime-valid inference. *Statist. Sci.* 38 (2023), 576–601.

[94] Y. Romano, E. Patterson, and E. Candès. 2020. Classification with valid and adaptive coverage. In *NeurIPS*.

[95] S. Sagawa, P. W. Koh, T. B. Hashimoto, and P. Liang. 2020. Distributionally robust neural networks for group shifts. In *ICLR*.

[96] J. M. Schwartz, M. George, S. C. Rossetti, P. C. Dykes, S. R. Minshall, E. Lucas, and K. D. Cato. 2023. Factors influencing clinician trust in predictive clinical decision support systems. *JAMIA Open* (2023).

[97] D. Sculley, G. Holt, D. Golovin, E. Davydov, T. Phillips, D. Ebner, V. Chaudhary, M. Young, J.-F. Crespo, and D. Dennison. 2015. Hidden technical debt in machine learning systems. *NeurIPS* (2015).

[98] L. Semenova, C. Rudin, and R. Parr. 2022. On the existence of simpler machine learning models. *ACM FAccT* (2022).

[99] L. Seyyed-Kalantari, H. Zhang, M. B. A. McDermott, I. Y. Chen, and M. Ghassemi. 2021. Underdiagnosis bias of artificial intelligence algorithms applied to chest radiographs in under-served patient populations. *Nature Medicine* 27 (2021), 2176–2182.

[100] G. Shafer. 2021. Testing by betting: a strategy for statistical and scientific communication. *Journal of the Royal Statistical Society A* (2021).

[101] K. Singh, A. Drouin, A. L. Beam, A. Igelnik, D. Agniel, M. J. Pencina, L. Wynants, R. D. Riley, G. S. Collins, and K. G. M. Moons. 2022. Continual evaluation for lifelong learning of clinical machine learning models. *npj Digital Medicine* 5 (2022).

[102] K. Stone, R. Zwiggelaar, P. Jones, and N. Mac Parthaláin. 2022. A systematic review of the prediction of hospital length of stay: towards a unified framework. *PLOS Digital Health* (2022).

[103] S. Tonekaboni, S. Joshi, M. D. McCradden, and A. Goldenberg. 2019. What clinicians want: contextualizing explainable machine learning for clinical end use. *Proceedings of Machine Learning Research (MLHC)* (2019).

[104] U.S. Food and Drug Administration. 2024. Predetermined change control plans for AI/ML-enabled device software functions. Final guidance, December 2024..

[105] B. Ustun, A. Spangher, and Y. Liu. 2019. Actionable recourse in linear classification. In *ACM FAccT*.

[106] B. Vasey, M. Nagendran, B. Campbell, D. A. Clifton, G. S. Collins, S. Denaxas, A. K. Denniston, L. Faes, B. Geerts, M. Ibrahim, et al. 2022. Reporting guideline for the early stage clinical evaluation of decision support systems driven by artificial intelligence: DECIDE-AI. *Nature Medicine* 28 (2022), 924–933.

[107] S. Verma and J. Rubin. 2018. Fairness definitions explained. In *FairWare at ICSE*.

[108] J. Ville. 1939. Étude critique de la notion de collectif. *Gauthier-Villars* (1939).

[109] D. A. Vyas, L. G. Eisenstein, and D. S. Jones. 2020. Hidden in plain sight: reconsidering the use of race correction in clinical algorithms. *New England Journal of Medicine* 383 (2020), 874–882.

[110] S. Wachter, B. Mittelstadt, and C. Russell. 2017. Counterfactual explanations without opening the black box. In *Harvard Journal of Law and Technology*.

[111] A. Wald. 1947. Sequential Analysis. *John Wiley & Sons* (1947).

[112] S. Wang, W. Guo, H. Narasimhan, A. Cotter, M. Gupta, and M. I. Jordan. 2020. Robust optimization for fairness with noisy protected groups. In *NeurIPS*.

[113] J. Watson-Daniels, D. C. Parkes, and B. Ustun. 2023. Predictive multiplicity in probabilistic classification. In *AAAI*.

[114] I. Waudby-Smith and A. Ramdas. 2024. Estimating means of bounded random variables by betting. *Journal of the Royal Statistical Society B* 86 (2024), 1–27.

[115] A. Wong, E. Otles, J. P. Donnelly, A. Krumm, J. McCullough, O. DeTroyer-Cooley, J. Pestrue, M. Phillips, J. Konye, C. Penoza, et al. 2021. External validation of a widely implemented proprietary sepsis prediction model in hospitalized patients. *JAMA Internal Medicine* 181 (2021), 1065–1070.

[116] Jacqueline G. You, Tina Hernandez-Boussard, Michael A. Pfeffer, Adam B. Landman, and Rebecca G. Mishuris. 2025. Clinical trials informed framework for real world clinical implementation and deployment of artificial intelligence applications. *npj Digital Medicine* (2025). Online February 17, 2025..

[117] R. V. Zicari, J. Brodersen, J. Brusseau, B. Dudder, T. Eichhorn, T. Ivanov, G. Kararigas, P. Kringen, M. McCullough, F. Moslein, N. Mushtaq, G. Roig, M. Stuermer, K. Tolle, J. J. Tithi, I. van Halem, and M. Westerlund. 2021. Z-Inspection: A Process to Assess Trustworthy AI. *IEEE Transactions on Technology and Society* 2, 2 (2021), 83–97.

A REPRODUCIBILITY APPENDIX

Figure / Section	n	seeds	Drift configuration
Sec. 6.2, Fig. 3	10 000	0 to 49	onset $t = 5\,000$
Sec. 6.3 Track 1, Fig. 4	1 000	0 to 29	none
Sec. 6.4, Figs. 6, 7	15 000	42	none
Sec. 6.5, Fig. 9	20 000	11; bootstraps 0 to 999	none
Sec. 6.6, Fig. 10	10 000	0 to 199 per α	none (H_0)
Sec. 6.7, Fig. 11	12 000	4	onset $t = 6\,000$
Sec. 6.11, Fig. 14	12 000	0 to 29 per ρ_1	onset $t = 6\,000$; $\rho_1 \in [0.06, 0.20]$
Sec. 6.12, Fig. 15	15 000	42	none

Table 10. Per-experiment synthesiser configuration. All experiments are reproducible via python -m chrono_fair.experiments.<expN>.